Frequentist, Bayesian and Machine Learning Analysis of the Matter–Antimatter Asymmetry in $B^{\pm} \to K^{\pm}K^{+}K^{-}$ Decays at LHCb

Final project for the *Statistics and Data Analysis* course
Academic Year 2025–2026

Giacomo Moretti and Marco Belotti
Università degli Studi di Milano-Bicocca

[TODO: Giacomo + Marco, write last. 3–5 sentences: dataset (LHCb Open Data, $B^{\pm} \to K^{\pm}K^{+}K^{-}$), the three methodological approaches compared (frequentist MLE/profile likelihood, Bayesian MCMC, ML-assisted selection), the headline result ($A_{\rm raw}$ estimate + consistency across methods), one sentence on what's new/interesting (e.g. goodness-of-fit model comparison, local Dalitz asymmetries). Don't give every number here – see §IX guidance.]

I. INTRODUCTION

Charge-parity (CP) violation – the small observed asymmetry between the behavior of matter and antimatter – is one of the necessary ingredients for the matter-dominated universe we observe, yet the CP violation built into the Standard Model is far too small to account for it. Studying CP violation in the decays of B mesons, where it is comparatively large and experimentally accessible, is one of the main ways to test the Standard Model's CKM description of the effect and search for physics beyond it. Charmless three-body decays such as $B^{\pm} \to h^{\pm}h^{+}h^{-}$ ($h = K, \pi$) are especially interesting in this respect: unlike a two-body decay, the three-body phase space (the Dalitz plot) allows the asymmetry to be studied not only as a single global number, but also *locally*, since interference between intermediate resonances can concentrate CP-violating effects in specific kinematic regions rather than spreading them uniformly [1, 2].

The quantity measured in this work is the raw asymmetry between the signal yields of the two charges,

$$A_{\rm raw} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}. \tag{1}$$

It differs from the physical asymmetry A_{CP} by the production asymmetry A_P of $B^{\pm}$ mesons in pp collisions and by the detection asymmetry A_D of the final state; to first order in these small quantities,

$$A_{CP} \approx A_{\rm raw} - A_P - A_D, \tag{2}$$

where A_D arises mainly from the different interaction cross sections of K^+ and K^- with the detector material and depends on the kaon momenta [1]. Neither correction is applied here, so $A_{\rm raw}$ is not a measurement of A_{CP}. The production asymmetry of $B^{\pm}$ mesons at the LHC is expected at the level of one per cent; in the published analysis of the same 2011 data, the two corrections, measured with $B^{\pm} \to J/\psi K^{\pm}$ decays, together with an acceptance correction, move the asymmetry of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ from $A_{\rm raw} = -0.060 \pm 0.007$ to $A_{CP} = -0.043 \pm 0.009\,(\text{stat}) \pm 0.003\,(\text{syst}) \pm 0.007\,(J/\psi K)$ [3].

In this work we use the LHCb Open Data record for $B^{\pm} \to K^{\pm}K^{+}K^{-}$ [4, 5], a subset of the 2011 data on which the first evidence of an inclusive CP asymmetry in this decay was published [3], to address the question: *to what extent do frequentist likelihood fits, Bayesian inference, and machine-learning-assisted event selection produce consistent, well-controlled estimates of the raw asymmetry $A_{\rm raw}$ in this decay?* The aim is not a competitive measurement of A_{CP}, but a methodological comparison of the three inference frameworks on the same data and the same physical question. Of the four channels in the record, $B^{\pm} \to K^{\pm}K^{+}K^{-}$ has the lowest background; the others ($\pi^{\pm}\pi^{+}\pi^{-}$, $K^{\pm}\pi^{+}\pi^{-}$, $\pi^{\pm}K^{+}K^{-}$) need a different particle-identification selection and a model of larger and peaking (misidentified) backgrounds, and are not analysed, to keep the focus on the comparison of the methods. The published A_{CP} cannot be reproduced from this record, which does not contain the control channel used to correct $A_{\rm raw}$ for the production and detection asymmetries (Sec. VIII); what we extend is the statistical treatment: a common model fitted with two inference methods, a common systematic study, a validation of the intervals, and a machine-learning selection whose effect on the measurement is tested. Section II describes the dataset and the preselection common to the three analyses; Secs. III and IV present the frequentist and Bayesian fits, Sec. V their common systematic study, and Sec. VI the machine-learning analysis; Sec. VII brings the three together.

II. DATA

A. Dataset and Preselection

The dataset is the CERN Open Data record 4900, "Matter Antimatter Differences – B meson decays to three hadrons" [4]: candidates for $B^{\pm} \to h^{\pm}h^{+}h^{-}$ decays collected by LHCb in 2011 at $\sqrt{s} = 7\,\text{TeV}$, reconstructed with the LHCb software and already passed through trigger and offline selection. The data are distributed as ROOT `TTrees` (`DecayTree`) containing only

the most relevant reconstructed branches, split across two files by magnet polarity (magnet up/down), for a combined 8,556,118 candidates. The same record also includes a simplified, detector-response-free phase-space simulation of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ (50,000 decays, half of each charge); it is used only as a sample without asymmetry to test the local-asymmetry procedure (Sec. IV H).

For each of the three final-state tracks ($H1$, $H2$, $H3$) the analysis reads momentum components (`PX`, `PY`, `PZ`), particle-identification (PID) likelihoods (`ProbK`, `ProbPi`), a muon flag (`isMuon`), and the track charge. The same preselection is used in all three analyses: a kaon-PID requirement and a muon veto on every track,

$$\texttt{H*_ProbPi} < 0.5 \ \wedge \ \texttt{H*_ProbK} > 0.5 \ \wedge \ \neg\texttt{H*_isMuon},$$

which retains 23,895 of the 8,556,118 candidates (0.28%), the PID requirement being by far the dominant reduction. The files contain no event identifier; as a proxy for multiple candidates from the same collision, only 42 of the candidates in the fit window (0.2%) share an identical track with another candidate, a negligible effect on the statistical uncertainties, and all are kept.

In this dataset only the charge combinations $(+1,-1,-1)$ and $(-1,+1,+1)$ occur for $(H1, H2, H3)$: $H1$ is always the kaon of charge opposite to the B, and $H2$, $H3$ are the interchangeable same-sign pair. The Dalitz-plot variables are therefore the invariant masses of the two opposite-sign pairs, $m^2(H1, H2)$ and $m^2(H1, H3)$.

B. Invariant Mass Reconstruction

The three final-state tracks are assigned the charged-kaon mass hypothesis ($M_K = 493.677$ MeV) regardless of their PID likelihoods, and their four-momenta are summed to reconstruct the B candidate, $m(KKK) = \sqrt{(\sum_i E_i)^2 - |\sum_i \vec{p}_i|^2}$. The nominal fit window, common to the frequentist and Bayesian analyses, is $[5100, 5600]$ MeV. Below $\sim$5150 MeV the background rises, consistent with a partially reconstructed contribution that a single exponential cannot describe; the window includes the start of this rise, which is described by a second exponential in the background model (Secs. III–IV). This leaves 19,731 preselected candidates in the fit window: 10,292 with B-charge $+1$ and 9439 with B-charge -1, split 11,636 (magnet down) / 8095 (magnet up). The machine-learning analysis uses the same window, with the region below 5150 MeV excluded from its background-like training sample (Sec. VI A).

C. Data Exploration

Figure I shows the reconstructed mass spectrum in the fit window, split by B charge (left) and by magnet polarity (right). The spectrum peaks at $m \approx 5284$ MeV over a falling combinatorial background, with a similar shape for the two charges and the two magnet polarities. The charge-counting asymmetry of all candidates in the fit window, signal and background together, is $A_{\text{count}} = (N^- - N^+)/(N^- + N^+) = -0.043 \pm 0.007$ (binomial uncertainty).

III. FREQUENTIST ANALYSIS

[TODO: Marco – intro, 1–2 sentences: unbinned extended maximum-likelihood fit with `iminuit`, profile likelihood and Wilks' theorem. Write in the past tense; call the asymmetry A_{raw} (not A_{CP}). Figures with `Report/figures/plot_style.py` (English, PDF).]

A. Mass Model and Fit Strategy

The frequentist fit uses the nominal mass model common to the two analyses (Sec. IV A): window $[5100, 5600]$ MeV, truncated Gaussian signal with shared μ, σ, double-exponential background with a shape and a yield for each charge, extended unbinned likelihood. It is maximized with `iminuit` [6], an interface to MINUIT [7] (`iminuit.cost.ExtendedUnbinnedNLL`), instead of being sampled by Markov chain Monte Carlo (MCMC). **[TODO: Marco – fitted yields and shapes, Hesse uncertainties, goodness of fit ($\chi^2 = 441.7/388$, 200 bins per charge), mass-fit figure with pulls; the sideband fit used for the starting values; the sideband-subtraction estimate of A_{raw} compared with the fit (-0.0503 ± 0.0083 in the re-run notebook).]**

B. Separate vs. Simultaneous Fit

In the *separate* fit the two charge samples are fitted independently and A_{raw} is computed from the two signal yields; in the *simultaneous* fit the parametrization $N_s^{\pm} = N_s(1 \mp A)/2$ estimates A directly, with mass, resolution and background parameters as nuisance parameters. **[TODO: Marco – both results (simultaneous -0.0441 ± 0.0085, separate -0.0442 ± 0.0087 in the notebook snapshot; re-check after rerun) and their agreement.]**

C. Profile Likelihood and Confidence Interval

The profile likelihood ratio,

$$q(A) = -2 \log \frac{L(A, \hat{\hat{\theta}})}{L(\hat{A}, \hat{\theta})},$$

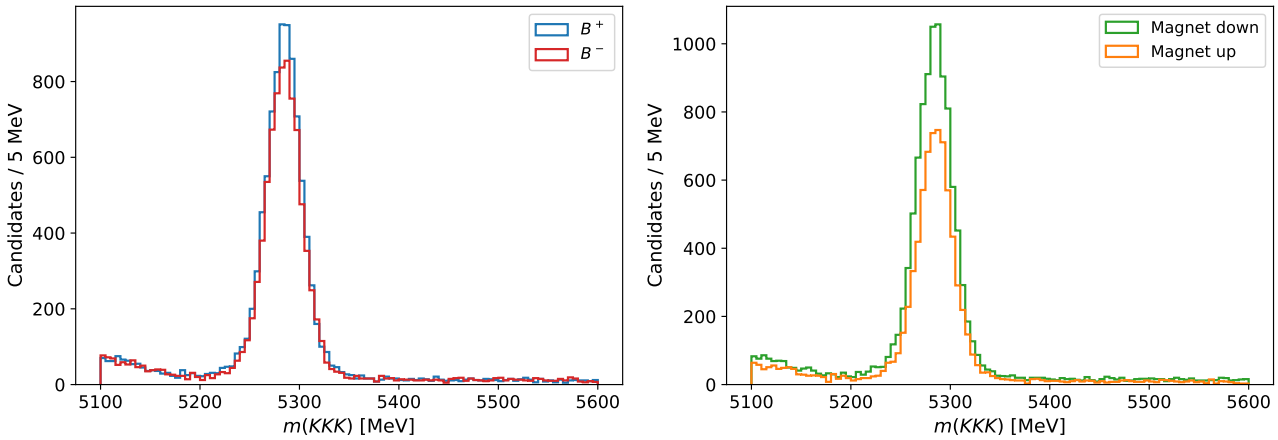


FIG. I. Reconstructed $B \to KKK$ invariant mass in the nominal fit window $[5100, 5600]$ MeV, split by B charge (left) and by magnet polarity (right). Both splits show a consistent peak at $m \approx 5284$ MeV over a falling combinatorial background, which rises below ∼5150 MeV.

gives a confidence interval on A via Wilks' theorem [8] and tests the null hypothesis $A = 0$ – the frequentist counterpart of the Savage-Dickey test of Sec. IV D. **[TODO: Marco – profile-likelihood figure; MLE, 68%/95% intervals and significance of $A \neq 0$ ($q(0) = 27.0$, 5.2σ with the nominal model); comparison with the Bayesian interval (-0.0443, $[-0.0528, -0.0358]$).]**

D. Coverage of the Wilks Intervals

[TODO: Marco – coverage of the profile-likelihood interval on A alone (1000 bootstrap replicas, nominal model, re-run notebook): $68.4 \pm 1.5\%$ for the 68.3% interval; describe the procedure and compare with the Bayesian coverage of Sec. IV G.]

E. Systematic Uncertainties

The systematic study is common to the frequentist and Bayesian fits and is presented in Sec. V. **[TODO: Marco – optional: the window-edge grid scan as a stability check (not a systematic; overlapping samples, Barlow). If the Dalitz region $[5, 10] \times [0, 5]$ GeV^2 is shown: it was chosen after looking at the map, so present it as exploratory, use $m^2_{\rm lo}/m^2_{\rm hi}$, and test against the global asymmetry.]**

IV. BAYESIAN ANALYSIS

A. Model and Priors

The mass spectrum of each charge sample is described by a signal-plus-background mixture on the nominal fit window $[5100, 5600]$ MeV (Sec. II B),

$$f^{\pm}(m) = N_s^{\pm}\, S(m; \mu, \sigma) + N_b^{\pm}\, B^{\pm}(m), \tag{3}$$

where S is a Gaussian truncated to the window, with mean μ and width σ shared by the two charges, and the background is a double exponential with a shape of its own for each charge,

$$B^{\pm}(m) = f_{\pm}\, E(m; \lambda_1^{\pm}) + (1 - f_{\pm})\, E(m; \lambda_2^{\pm}),$$

with $E(m; \lambda) \propto e^{-\lambda m}$ a truncated exponential normalized on the window. The steep component (λ_1) describes the rise of the background towards the lower edge of the window, the slow one (λ_2) the combinatorial background under and above the peak. The mixture is invariant under the exchange $(\lambda_1, f) \leftrightarrow (\lambda_2, 1 - f)$, which would give the posterior two identical modes (label switching); the ordering $\lambda_1 > \lambda_2$ removes this symmetry. Signal and background yields are split between the charges as

$$N_s^{\pm} = \tfrac{N_s}{2}(1 \mp A), \qquad N_b^{\pm} = \tfrac{N_b}{2}(1 \mp A_b),$$

so that $A = (N_s^- - N_s^+)/(N_s^- + N_s^+)$, the raw asymmetry $A_{\rm raw}$ of Eq. (1), is a fit parameter rather than being derived from two independent yields, and the background asymmetry A_b is free. The parameter vector has twelve components, $\theta = (N_s, N_b, A, A_b, \mu, \sigma, \lambda_1^{\pm}, \lambda_2^{\pm}, f_{\pm})$.

Given the data $D = \{(m_i, q_i)\}_{i=1}^{N}$ (mass and charge of each candidate), Bayes' theorem gives the posterior

$$p(\theta \mid D) = \frac{\mathcal{L}(D \mid \theta)\, \pi(\theta)}{p(D)}, \tag{4}$$

with the extended unbinned likelihood of the two charge samples

$$\mathcal{L}(D \mid \theta) \propto e^{-(N_s+N_b)} \prod_{i:\, q_i=+1} \Big[N_s^+ S(m_i)+N_b^+ B^+(m_i)\Big] \times \prod_{i:\, q_i=-1} \Big[N_s^- S(m_i) + N_b^- B^-(m_i)\Big], \quad (5)$$

where $S(m_i) \equiv S(m_i;\mu,\sigma)$. The priors are independent and uniform: $N_s, N_b \in (0, 5N)$ with $N = 19{,}731$, $A, A_b \in (-1, 1)$, $\mu \in (5260, 5310)$ MeV, $\sigma \in (3, 40)$ MeV, $\lambda_1^{\pm} \in (10^{-5}, 0.1)$ MeV^{-1}, $\lambda_2^{\pm} \in (10^{-5}, \lambda_1^{\pm})$, $f_{\pm} \in (0, 1)$. The prior density is therefore a constant on the allowed region Θ,

$$\pi(\theta) \propto \mathbf{1}\,[\theta \in \Theta]\,, \quad (6)$$

and inside it the posterior is proportional to the likelihood. A flat prior is not uninformative in an absolute sense, since it is not invariant under reparametrization (a prior uniform in λ is not uniform in $1/\lambda$); with $\sim 2 \times 10^4$ events, however, the posterior is dominated by the likelihood, as the prior-sensitivity check of Sec. IV F confirms for A. The evidence $p(D)$ is a constant normalization: the MCMC samples $\mathcal{L}\,\pi$ without computing it, and the Savage-Dickey ratio of Sec. IV D avoids computing it as well. The marginal posterior of the asymmetry, $p(A \mid D) = \int p(\theta \mid D)\, d\theta_{\neq A}$, is the histogram of the A component of the chain.

B. MCMC Sampling and Convergence Diagnostics

The posterior was sampled with the affine-invariant ensemble sampler of Goodman and Weare [9] as implemented in `emcee` [10]. The maximum of the posterior was first located numerically (Nelder-Mead), and 32 walkers were initialized in a narrow ball around it and run for 25,000 steps, with the first 4166 (1/6) discarded as burn-in, leaving 6.7×10^5 samples; the mean acceptance fraction is 0.38. Convergence was checked through the integrated autocorrelation time of each parameter, $\tau = 1 + 2\sum_{t\geq 1} \rho(t)$, with $\rho(t)$ the autocorrelation of the chain at lag t. We find $\tau \in [147, 183]$ steps, the longest for the background-shape parameters: the post-burn-in chain is $N/\tau \geq 114$ autocorrelation times long for every parameter, above the threshold of 50 recommended for a reliable estimate of τ [10], and corresponds to $\gtrsim 3600$ effectively independent samples per parameter. To check that the chain is not trapped near its starting point, the fit was repeated twice with the walkers initialized uniformly over the whole prior box and the first half of the chain discarded: both runs converge to the same single mode, with $A_{\mathrm{raw}} = -0.0446$ and -0.0442, the same intervals and the same values of all other parameters as the nominal fit. The trace plot of such a run (Fig. XX, Appendix A) shows all walkers reaching the bulk of the posterior within $\sim$2000 steps, the background fraction f_+ last, well inside the discarded half of the chain.

C. Posterior Results and Credible Intervals

The marginal posterior on the raw asymmetry gives (median and equal-tailed credible intervals, CI)

$$A_{\mathrm{raw}} = -0.0443,$$
$$68\%\ \mathrm{CI}\ [-0.0528, -0.0358],$$
$$95\%\ \mathrm{CI}\ [-0.0608, -0.0276].$$

The posterior is very close to Gaussian, with standard deviation 0.0085; its 68% interval is symmetric about the median to within 2×10^{-5}, so the equal-tailed intervals coincide with the highest-posterior-density ones at the quoted precision. Since the priors are uniform, the posterior mode (the maximum a posteriori, MAP, estimate, -0.0441) coincides with the maximum-likelihood estimate, so this result can be compared directly with the frequentist fit of the same model (Sec. III). The published analysis of the same 2011 data, with its own selection (which includes a D^0 veto) and a mass model with partially reconstructed and misidentified backgrounds, finds $A_{\mathrm{raw}} = -0.060 \pm 0.007$ [3]; with the D^0 veto our fit gives -0.0496 ± 0.0088 (Sec. IV F), compatible within the statistical uncertainties, although the two selections overlap and differ. After the corrections of Eq. (2), the published asymmetry is $A_{CP} = -0.043 \pm 0.009 \pm 0.003 \pm 0.007$ [3], and the most precise later measurement, on a larger data set, is $A_{CP} = -0.037 \pm 0.002 \pm 0.002 \pm 0.003$ [1, 2]. The other physics parameters are $N_s = 15{,}328 \pm 133$, $N_b = 4404 \pm 82$, $A_b = -0.040 \pm 0.018$, $\mu = 5284.39 \pm 0.16$ MeV and $\sigma = 18.00 \pm 0.14$ MeV (Fig. II). A is almost uncorrelated with the other parameters: its largest correlation, -0.17, is with A_b.

D. Bayes Factors

Computing the evidence of each hypothesis by direct numerical integration is impractical for a 12-parameter posterior. When the null hypothesis fixes one parameter, H_0: $A = 0$, nested in H_1 (A free), and the priors on the other parameters are shared, the Bayes factor reduces to the Savage-Dickey density ratio [11],

$$BF_{01} = \frac{p(A = 0 \mid D)}{\pi(A = 0)}, \qquad \pi(A = 0) = \tfrac{1}{2}.$$

With equal prior probabilities for the two hypotheses, BF_{01} is also the posterior odds of H_0 against H_1. $A = 0$ lies 5.2 posterior standard deviations from the median, in a tail where the chain contains essentially no samples, so a kernel density estimate of $p(A = 0 \mid D)$ would only reflect the kernel tails of the few most extreme samples. We therefore use the Gaussian approximation of the marginal posterior (Sec. IV C), with the mean and standard deviation of the chain, which gives $BF_{01} \approx 1.1 \times 10^{-4}$ ($\ln BF_{10} \approx 9.1$): decisive evidence against $A = 0$ on Jeffreys' scale [12] ($BF_{10} > 100$). As for any Bayes factor

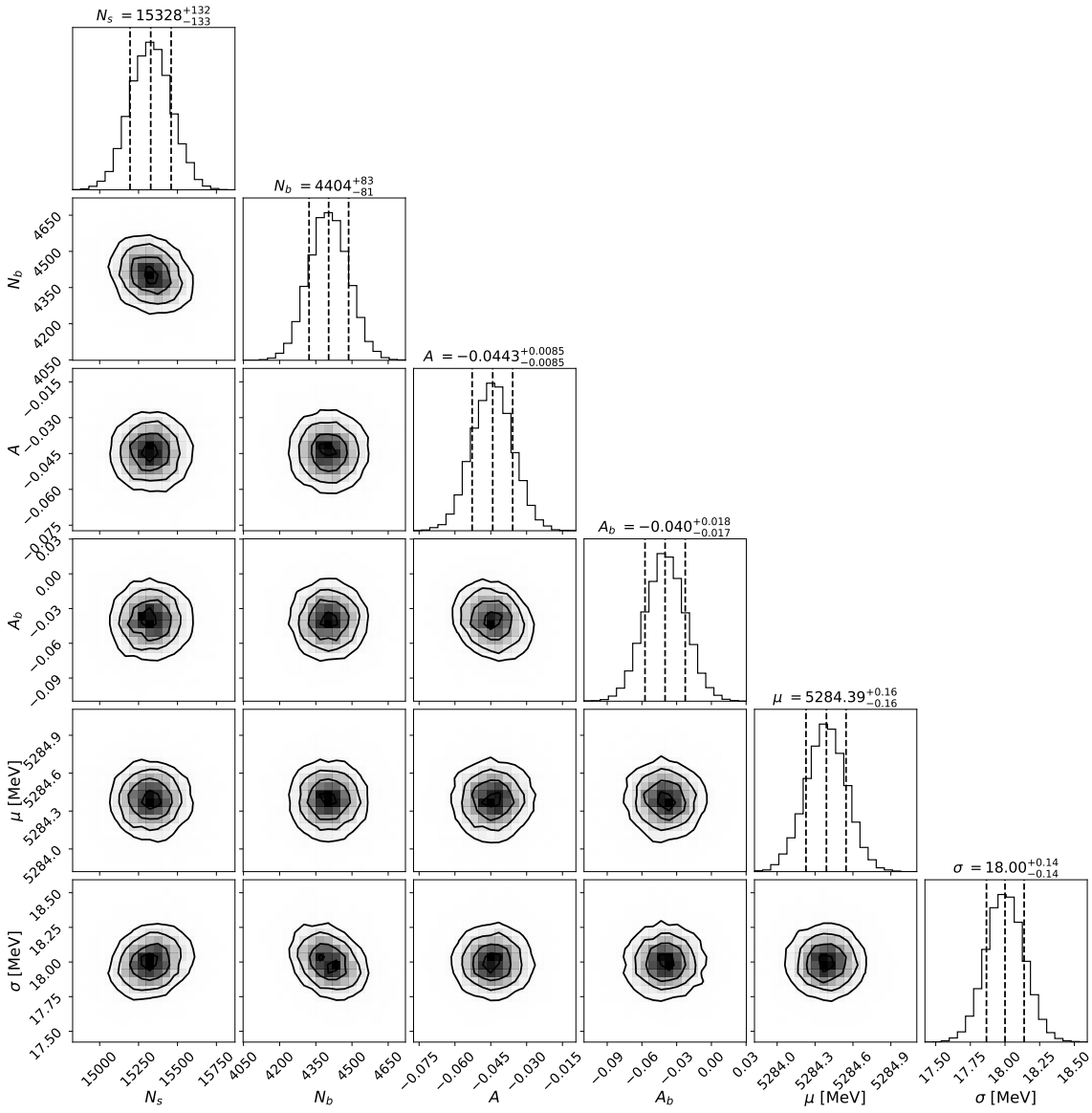


FIG. II. Posterior of the six physics parameters of the nominal fit (the six background-shape parameters are marginalized). Diagonal: marginal posteriors with the 16th, 50th and 84th percentiles (dashed lines and titles). Off-diagonal: 2D marginals, contours at the $0.5, 1, 1.5, 2\sigma$ levels of a 2D Gaussian.

of a point null against a free parameter, the value scales with the prior width – the Ockham factor built into the Bayes factor, which penalizes a free parameter in proportion to the prior volume it is allowed to explore, and the mechanism behind Lindley's paradox: with the narrower prior $A \in (-0.3, 0.3)$ of Sec. IV F, $\pi(A = 0)$ grows to $1/0.6$ and BF_{01} drops by a factor 3.3, which leaves the conclusion unchanged. Figure III shows the two densities of the ratio.

The same test on the background asymmetry shows the Ockham factor at work. $A_b = -0.040 \pm 0.018$ lies 2.3 standard deviations from zero, yet the Savage-Dickey ratio gives $BF_{01}(A_b = 0) \approx 3.5$: the data mildly *favor* a charge-symmetric background, because the gain in likelihood does not compensate for spreading the prior of A_b over $(-1, 1)$. A negative background asymmetry is plausible: besides combinatorial background, the sample contains misidentified $B^\pm \to K^\pm\pi^+\pi^-$ and $B^\pm \to \pi^\pm K^+K^-$ decays, with A_{CP} values of their own, whose contribution grows when the kaon identification is loosened (Sec. V). A_b is nevertheless kept free in the nominal model, which avoids imposing an assumption on the background that the frequentist analysis does not make; fixing $A_b = 0$ is one of the variants of the systematic study (Sec. V).

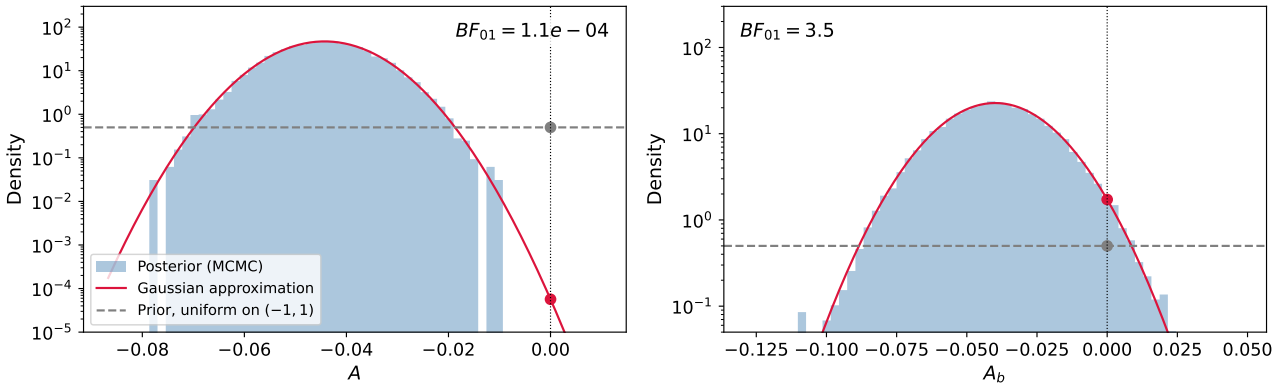


FIG. III. Savage-Dickey ratio for A (left) and A_b (right): marginal posterior (histogram of the chain and Gaussian approximation) and prior density, logarithmic scale. The Bayes factor BF_{01} is the ratio of the two densities at zero (dots). For A the posterior density at zero is 10^{-4} times the prior one; for A_b, 2.3 standard deviations from zero, it is higher than the prior density, and the data favor $A_b = 0$.

E. Goodness of Fit and Model Choice

A binned comparison of data and posterior-median model ($\chi^2 = \sum(\text{obs} - \text{exp})^2/\text{exp}$ over 50 bins of 10 MeV per charge, $100 - 12 = 88$ degrees of freedom) gives $\chi^2 = 164.1/88$ ($p = 1.6\times10^{-6}$), with 95.4 for B^+ and 68.7 for B^-. The largest pulls, in the B^+ sample, sit on the two flanks of the peak, about $\pm3\sigma$ from μ (Fig. IV): the observed signal shape has heavier tails than a Gaussian. Replacing the Gaussian by a Crystal Ball shape [13], a Gaussian core with a power-law low-mass tail,

$$S_{\text{CB}}(x) \propto \begin{cases} e^{-x^2/2}, & x > -\beta, \\ \left(\frac{\kappa}{\beta}\right)^{\kappa} e^{-\beta^2/2} \left(\frac{\kappa}{\beta} - \beta - x\right)^{-\kappa}, & x \le -\beta, \end{cases} \tag{7}$$

with $x = (m-\mu)/\sigma$, β the point where the tail starts (in units of σ) and κ its power, both free, improves the fit to $\chi^2 = 131.0/86$ ($p = 1.3\times10^{-3}$, $\beta = 1.71$, $\kappa = 35$) but moves A_{raw} by only -0.0002 (Table I). The asymmetry does not depend on the signal shape even though none of the models reaches $\chi^2/\text{ndof} \approx 1$. The Gaussian is nevertheless kept as the nominal signal model, common to the two analyses: the Crystal Ball adds two tail parameters that are poorly constrained (its power κ runs to large values), and the effect of the signal shape on A_{raw} is covered by the systematic study (Sec. V).

The background model is constrained by the data much more strongly. A single exponential per charge is rejected ($\chi^2 = 297.7/92$, $\Delta\chi^2 = 134$ for four fewer parameters), and so it is not used even as a systematic variant (Sec. V). A shape shared between the two charges ($\Delta\chi^2 = 9.9$ for three fewer parameters) and a charge-symmetric background ($\Delta\chi^2 = 5.7$ for one fewer) are acceptable descriptions and shift A_{raw} by -0.0037 and -0.0028: the background treatment is the largest model dependence of the fit. The lower edge of the window is a compromise between two failures. Extending the window to 5050 MeV degrades the fit ($\chi^2 = 217.4/98$), because the double exponential no longer follows the background further below the peak; raising it to 5150 MeV removes most of the region that constrains the steep component, which becomes poorly identified – the chain for its fraction f_+ mixes slowly ($N/\tau = 10$) – while A_{raw} itself is unchanged (-0.0444).

The binned χ^2 uses the posterior medians only. A posterior predictive check [14] accounts for the posterior uncertainty and tests specific features of the data: for 1000 parameter sets θ_j drawn from the posterior, a dataset y_j^{rep} is generated from the model, and a discrepancy $T(y,\theta_j)$ is computed both on the replica and on the data; the posterior predictive p-value is $p_B = P\big(T(y^{\text{rep}},\theta) \ge T(y,\theta) \mid y\big)$. For the binned χ^2 the replicas give 104 ± 15 and the data 175 (median over θ_j): none of the replicas reaches the observed value ($p_B < 10^{-3}$), with the misfit concentrated in the B^+ sample ($p_B < 10^{-3}$, against 0.03 for B^-). The discrepancies that probe the charge structure of the model are instead reproduced: the counting asymmetry of the whole window ($p_B = 0.996$), of the signal window (0.99) and of the sidebands (0.73) (Fig. V). The model misdescribes the shape of the spectrum, mainly the signal tails, but not the split between the charges that determines A_{raw}, consistently with the stability of A_{raw} under changes of the signal model.

Model	A_{raw}	χ^2/ndof	p
Nominal	-0.0443	164.1/88	2×10^{-6}
Crystal Ball signal	-0.0445	131.0/86	1×10^{-3}
μ, σ per charge	-0.0443	164.2/86	8×10^{-7}
Shared bkg. shape	-0.0480	174.1/91	4×10^{-7}
Symmetric bkg. ($A_b = 0$)	-0.0471	169.8/89	5×10^{-7}
Single exponential	-0.0480	297.7/92	2×10^{-23}

TABLE I. A_{raw} (posterior median) and goodness of fit of the nominal model (Gaussian signal, double exponential per charge) and of the alternative signal and background models, same data and window.

F. Cross-Checks

The fit was repeated separately on the two magnet polarities, whose left-right detection asymmetries are opposite and partially cancel in the combined sample: $A_{\rm raw}(\text{down}) = -0.0497 \pm 0.0112$ ($n = 11{,}636$) and $A_{\rm raw}(\text{up}) = -0.0378 \pm 0.0129$ ($n = 8095$). They differ by 0.7σ, so there is no indication that the asymmetry is an artifact of one polarity (Fig. VI). A narrower prior, $A \in (-0.3, 0.3)$, leaves the posterior unchanged ($A_{\rm raw} = -0.0443$), as expected for a posterior far from the prior bounds; the prior width matters only for the Bayes factor (Sec. IV D).

The sample contains $B^\pm$ decays through intermediate charm states, $B^\pm \to D^0 K^\pm$, $J/\psi K^\pm$ and $\chi_{c0} K^\pm$ with D^0, J/ψ, $\chi_{c0} \to K^+K^-$, which are visible as peaks in the opposite-sign $m(K^+K^-)$ spectra. They are part of the $B^\pm \to K^\pm K^+ K^-$ final state and are kept in the nominal sample; removing them changes the observable, from the asymmetry of all $B^\pm \to K^\pm K^+ K^-$ decays to that of the charmless ones, so the vetoes are reported as an informative check rather than as a systematic variation. A candidate is vetoed if either opposite-sign pair, $m(H1, H2)$ or $m(H1, H3)$, lies within ± 30 MeV of the D^0 or J/ψ mass or within ± 60 MeV of the χ_{c0} mass (world-average masses; the wider window for the χ_{c0} reflects its natural width). The D^0 veto alone keeps 18,744 candidates and gives $A_{\rm raw} = -0.0496 \pm 0.0088$; the three vetoes together keep 16,367 and give -0.0423 ± 0.0094. The shifts with respect to the nominal fit, -0.0053 and $+0.0020$, compare with expected statistical fluctuations of 0.0023 and 0.0040 for a subsample of the nominal one (Sec. V); the D^0 veto moves $A_{\rm raw}$ by 2.3 times this amount, i.e. the vetoed $D^0 K^\pm$ candidates have a less negative asymmetry than the rest of the sample.

G. Toy-MC Coverage Validation

A credible interval states where the parameter lies given the data and the prior; nothing guarantees that it also has frequentist coverage, i.e. that over repeated experiments it contains the true value with the stated probability. To check this, 200 pseudo-datasets were generated from the fitted model (parametric bootstrap, with the posterior medians as ground truth and Poisson-fluctuated yields for each charge and component; the truncated Gaussian and exponential masses are drawn with the transformation method, i.e. by inverting their cumulative distributions) and each was refit with the nominal procedure and a 12,000-step chain. The fraction of replicas whose interval contains the injected A is 0.650 ± 0.033 for the 68% CI and 0.930 ± 0.015 for the 95% CI (binomial uncertainties), 1.0σ and 1.3σ below nominal: the coverage is consistent with nominal. The estimates themselves are unbiased and their spread matches the reported uncertainty (Fig. VII): their mean differs from the injected value by -0.0002 ± 0.0006, their standard deviation, 0.0088, agrees with the posterior standard deviation of the data fit, 0.0085, and the pulls have mean -0.03 ± 0.07 and standard deviation 1.04.

The toy chains are shorter than the nominal one, and in 124 of the 200 replicas the slowest-mixing parameter, always a background-shape parameter, has $N/\tau < 50$ (median 48, minimum 22). The criterion on the slowest parameter is conservative for the quantity tested here: for A itself the median autocorrelation time is 156 steps, so each chain is at least 45 autocorrelation times long in A and provides $\gtrsim 1400$ effectively independent samples of it. The Monte Carlo error on the percentiles of A is then a few per cent of $\sigma(A)$, negligible against the binomial uncertainty of the coverage, and the replicas with $N/\tau < 50$ do not show a lower coverage than the others. This is a closure test of the inference procedure: since the toys are generated from the same model that is fitted, it does not probe the model mismatch of Sec. IV E, which is instead addressed by the systematic study.

H. Local Dalitz-Plot Asymmetry (Fixed Binning)

To test whether the asymmetry is uniform over phase space or concentrated in specific regions, each candidate was placed in the Dalitz plane using the two opposite-sign K^+K^- invariant masses $m^2(H1, H2)$ and $m^2(H1, H3)$ (Sec. II A), sorted into $m^2_{\rm lo}$ and $m^2_{\rm hi}$ to remove the $H2/H3$ labeling ambiguity. The plane $[0, 16] \times [0, 30]$ GeV^2 in $(m^2_{\rm lo}, m^2_{\rm hi})$ was divided into an 8×8 grid, fixed before looking at the asymmetries. In each bin with at least 20 candidates (32 of 64), the local asymmetry $A_i = (N_i^- - N_i^+)/N_i$, with binomial uncertainty $\sigma(A_i) = \sqrt{(1 - A_i^2)/N_i}$, was compared with the global counting asymmetry $A_{\rm glob} = -0.043$ through $S_i = (A_i - A_{\rm glob})/\sigma(A_i)$; a comparison with zero would instead mostly reflect the global asymmetry itself.

Two of the 32 bins have $|S_i| > 3$, adjacent to each other at $m^2_{\rm lo} < 4$ GeV^2 and $7.5 < m^2_{\rm hi} < 11.25$ GeV^2 ($S_i = -3.7$ and -3.0), with A_i more negative than the global value; the bin below the first one ($m^2_{\rm lo} < 2$, $3.75 < m^2_{\rm hi} < 7.5$ GeV^2) has $S_i = -2.98$, just under the threshold (Fig. VIII); about 0.09 would be expected if the asymmetry were uniform, with a probability of 3×10^{-3} for two or more. The largest positive deviations, close to $+3$, lie at high $m^2_{\rm hi}$. A global test of homogeneity, $\chi^2 = \sum_i S_i^2 = 80.8$ for 31 degrees of freedom ($p = 3 \times 10^{-6}$), confirms that the local asymmetry is not uniform. Restricting the sample to the signal window ($|m - \mu| < 2\sigma$, $n = 15{,}130$, where the fitted model gives a signal purity of $\approx$96%) gives the same picture – three bins with $|S_i| > 3$, $\chi^2 = 95.6/28$ – so the structure is not produced by the combinatorial background. Nor is it produced by the charm contributions: with the D^0 veto two bins still exceed 3σ ($\chi^2 = 79.2/31$), and with all three vetoes one remains ($\chi^2 = 74.4/31$, $p = 2 \times 10^{-5}$). It does not depend on the binning either: 6×6, 10×10 and 12×12 grids give $\chi^2 = 81.0/19$, $97.1/43$ and $120.0/60$

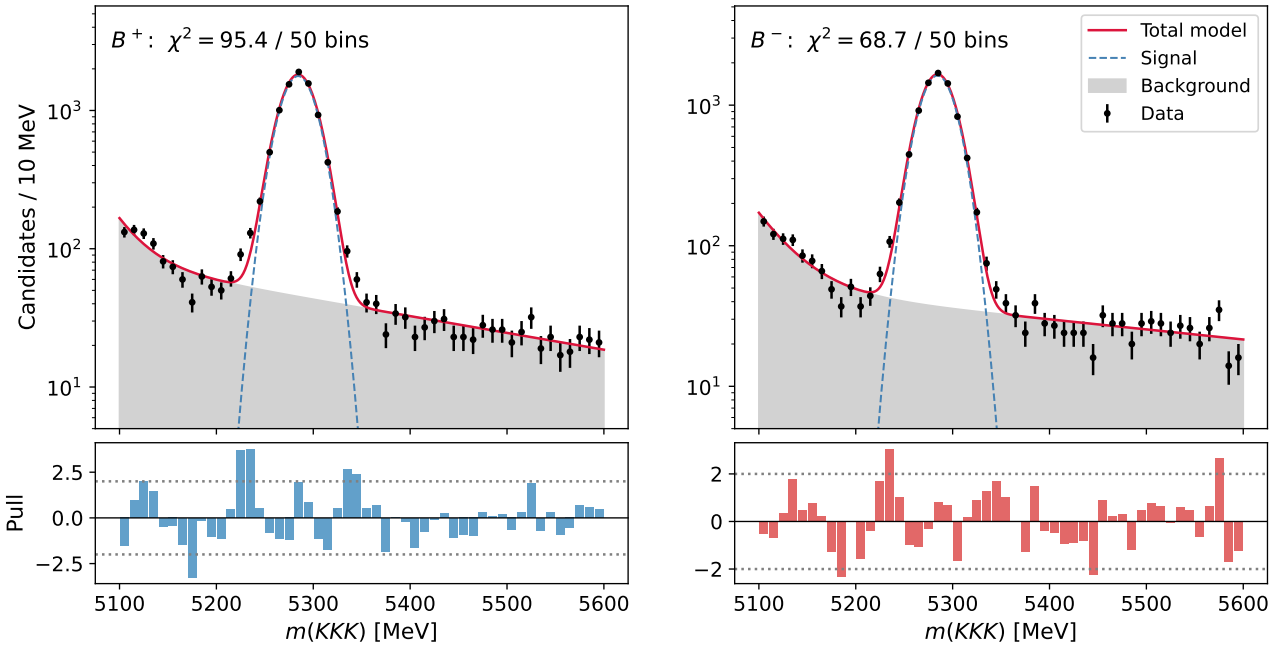


FIG. IV. Mass distributions of B^+ (left) and B^- (right) candidates with the nominal model (posterior medians, logarithmic scale): total, signal and background components. Bottom: pulls, $(\text{obs} - \text{exp})/\sqrt{\text{exp}}$.

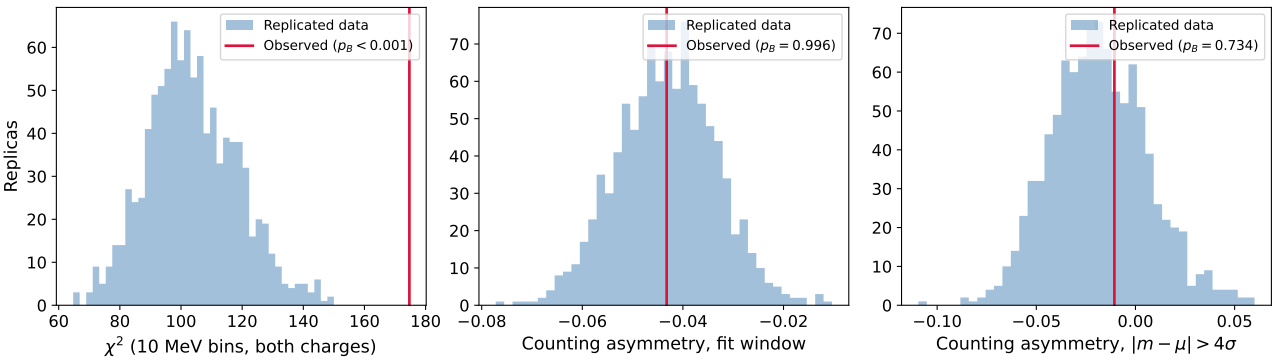


FIG. V. Posterior predictive check: distributions of three discrepancies over 1000 datasets replicated from posterior draws, compared with the observed values. Left: binned χ^2; center and right: counting asymmetry of the fit window and of the sidebands.

($p \leq 7 \times 10^{-6}$).

Two null tests check the procedure itself. Shuffling the charges among the candidates removes any local asymmetry while keeping the occupancy of the bins and the global asymmetry of the data: over 2000 permutations the 8×8 grid gives $\chi^2 = 31.4 \pm 8.1$, as expected for 31 degrees of freedom, and no permutation reaches the observed value (permutation $p < 5 \times 10^{-4}$; $p = 1 \times 10^{-3}$ for the 12×12 grid). The phase-space simulation of the record, without asymmetry by construction, gives $\chi^2 = 40.0/30$ ($p = 0.11$) with the same procedure. The homogeneity test therefore has its nominal behavior, and the non-uniformity is a property of the data.

The structure does not depend on the magnet polarity either. The left-right part of the detection asymmetry changes sign when the polarity is reversed, so a local asymmetry produced by it would appear with opposite deviations in the two samples. In the region of the two significant bins ($m^2_{\text{lo}} < 4$ GeV^2, $7.5 < m^2_{\text{hi}} < 11.25$ GeV^2) the counting asymmetry is -0.129 ± 0.023 with the magnet down and -0.126 ± 0.028 with the magnet up, although the global asymmetries of the two samples differ (-0.051 and -0.032), and the two maps agree bin by bin ($\chi^2 = 23.0$ for 29 bins, $p = 0.78$). This split does not test the part of the detection asymmetry that does not change sign with the polarity, the different interaction of K^+ and K^- with the detector material.

The ordered Dalitz plot (Fig. IX, left) shows the two-body resonances of the decay: the narrow $\phi(1020)$ band

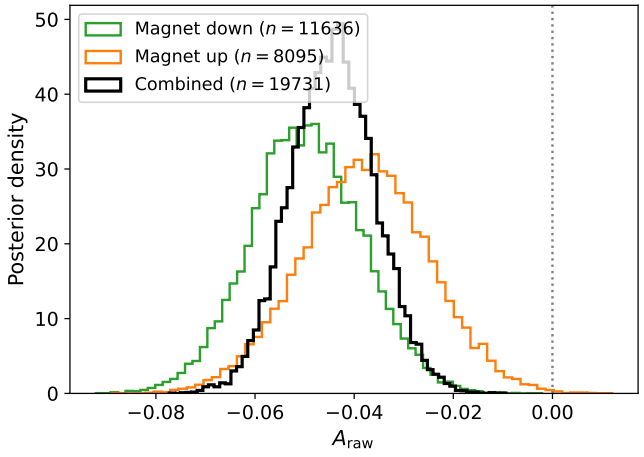


FIG. VI. Posterior of A_{raw} for the magnet-down, magnet-up and combined samples.

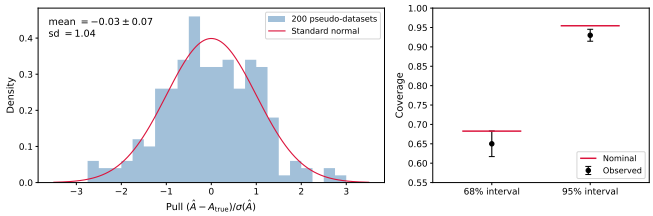


FIG. VII. Toy-MC validation, 200 pseudo-datasets. Left: pulls $(\hat{A} - A_{\mathrm{true}})/\sigma(\hat{A})$, with $\hat{A}$ and $\sigma(\hat{A})$ the posterior median and standard deviation of each replica, compared with a standard normal. Right: fraction of replicas whose 68% and 95% credible intervals contain the injected value, with binomial uncertainties, and the nominal coverage.

at the kinematic edge, $m^2_{\mathrm{lo}} \approx 1.04$ GeV^2, and the charm bands of the D^0, J/ψ and χ_{c0}. The negative local asymmetry is not produced by the $\phi(1020)$: its band ($|m_{\mathrm{lo}} - m_\phi| < 10$ MeV, 1964 candidates) has $A_{\mathrm{count}} = -0.004 \pm 0.023$, while the significant bins without it have -0.145 ± 0.019. The asymmetry sits in the low-mass K^+K^- region just above the ϕ, which is where the published analysis of the same data located it, defining the region $1.2 < m^2_{\mathrm{lo}} < 2.0$ GeV^2, $m^2_{\mathrm{hi}} < 15$ GeV^2 and measuring there $A_{\mathrm{raw}} = -0.239 \pm 0.020$ [3]. Since that region is fixed by the paper and not chosen on our data, it allows an unbiased comparison: fitting the 2146 candidates in it with the nominal model gives $A_{\mathrm{raw}} = -0.234 \pm 0.025$ (Bayesian; the chain mixes slowly in the poorly constrained second background exponential, $N/\tau = 15$) and -0.237 ± 0.025 (frequentist, 9.4σ from zero), reproducing the published value; the mass spectra of the two charges in the region (Fig. IX, right) show the difference directly. Three caveats apply: the binning is coarse and was not optimized; the asymmetries are computed on all candidates, without the sWeight-based background subtraction that a full analysis would use (Sec. VIII); and, since A_{raw} is not corrected for detection asymmetries, local variations can still be partly instrumental – the K^+/K^- interaction asymmetry depends on the kaon momenta, which are correlated with the position in the Dalitz plane, and is not removed by the polarity split. Section VI E revisits the same question with an adaptive, data-driven partition (K-means).

I. Bayesian and Frequentist Intervals at Low Statistics

With $\sim 2 \times 10^4$ candidates the posterior is Gaussian and, with flat priors, the Bayesian and frequentist results coincide by construction (Sec. IV C). To find where they could differ, the same likelihood was analysed with both methods on pseudo-datasets of $N = 50$, 150, 500 and 2000 candidates, 300 for each size. The model is the six-parameter one of Sec. IV J (Gaussian signal, one exponential background shared by the charges), which stays identifiable down to a few tens of events; the parameters are those of its fit to the data, with the yields rescaled. Each replica is analysed with the MCMC (median and equal-tailed credible intervals) and by maximum likelihood with `iminuit` (Hesse intervals, and Minos intervals from the profile likelihood and Wilks' theorem).

The two methods agree at every size (Fig. X). The coverages of the three intervals are compatible with nominal and with each other, the posterior median and the maximum-likelihood estimate differ on average by 5% of the statistical uncertainty at $N = 50$ and by less above, and the credible intervals stay symmetric. The asymmetry is a ratio of Poisson yields, whose likelihood is close to Gaussian already with a few tens of signal events: in this problem flat-prior credible intervals and profile-likelihood confidence intervals give the same answer, and Wilks' theorem holds, down to $N = 50$. The two approaches differ in what the interval means and in hypothesis testing (Sec. IV D), not in the numbers. At $N = 2000$ all three intervals over-cover slightly (0.98–0.99 for the 95% interval); since the effect is the same for the three methods, it is a feature of this set of replicas rather than of either inference.

J. A Minimal Model as Cross-Check

Before the nominal model was fixed, the same analysis was carried out with a minimal six-parameter model: a narrower window, [5150, 5450] MeV, which excludes most of the rising background below the peak, a single exponential background with a shape and a yield shared by the two charges, and the same Gaussian signal. On its 17,839 candidates it gives $A_{\mathrm{raw}} = -0.0497$ (68% CI $[-0.058, -0.041]$) with $\chi^2/\mathrm{ndof} = 127.4/54$, and -0.0457 when the background yields of the two charges are left free. Its toy validation, with 500 pseudo-datasets, gives coverage consistent with nominal (0.664 and 0.962 for the 68% and 95% intervals). The difference from the nomi-

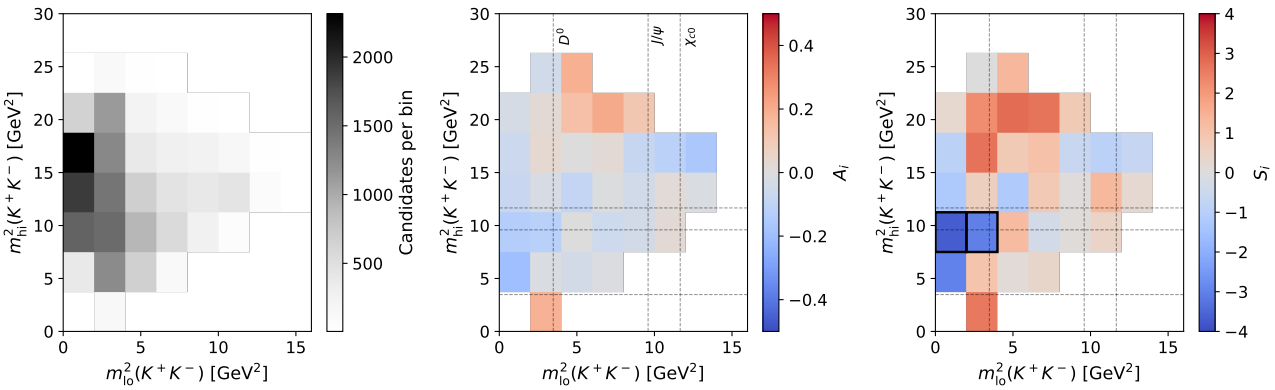


FIG. VIII. Fixed 8×8 Dalitz-plot binning, all candidates in the fit window: number of candidates per bin (left), local asymmetry A_i for bins with ≥ 20 candidates (center), and local significance with respect to the global asymmetry, $(A_i - A_{\rm glob})/\sigma(A_i)$ (right). The two bins with $|S_i| > 3$ are outlined. Dashed lines: the squared masses of the D^0, J/ψ and χ_{c0}, in either opposite-sign pair.

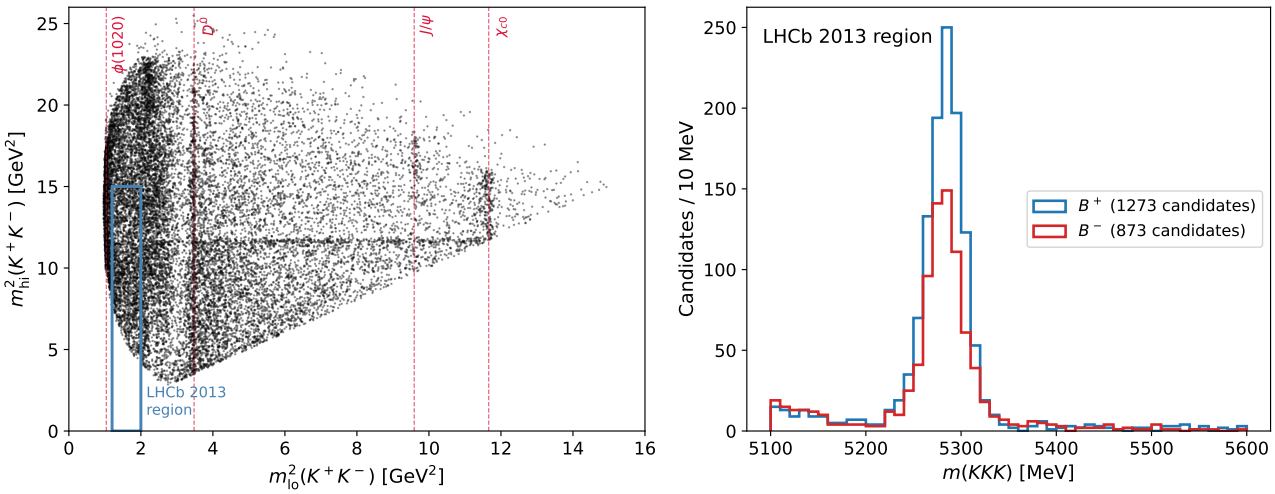


FIG. IX. Left: ordered Dalitz plot of the candidates in the fit window, with the squared masses of the $\phi(1020)$, D^0, J/ψ and χ_{c0} (dashed) and the region of the published analysis of the same data [3] (box). Right: mass spectra of B^+ and B^- candidates in that region, same binning: the B^- peak is visibly smaller.

nal result, 0.005, is driven by the background treatment, as in the systematic study of Sec. V, and is smaller than the statistical uncertainty.

V. SYSTEMATIC UNCERTAINTIES

The frequentist and Bayesian fits share one systematic study: the same thirteen variants of the nominal choices, in four groups, and the same rule to combine them. Each variant is fitted by both methods on exactly the same candidates.

- *Window*: lower edge at 5050 or 5150 MeV, upper edge at 5450 or 5700 MeV.
- *Background*: a single exponential per charge; a double exponential with the shape shared between charges; a charge-symmetric background ($A_b = 0$).
- *Signal*: a Crystal Ball shape with free tail parameters, Eq. (7); μ and σ separate for the two charges.
- *Preselection*: one cut at a time loosened or tightened, `ProbK` > 0.4 or > 0.6, `ProbPi` < 0.4 or < 0.6, on all three tracks.

The shift $\Delta = A^{\rm var}_{\rm raw} - A^{\rm nom}_{\rm raw}$ of a variant is not necessarily a systematic effect. When a variant changes the sample (window, preselection), the two samples are nested and part of the shift is a statistical fluctuation, with expected size $\sigma_B = \sqrt{|\sigma^2_{\rm var} - \sigma^2_{\rm nom}|}$, σ being the statistical uncertainty of each fit [15]; such a variant contributes only if $|\Delta| > \sigma_B$. A variant that the data reject, with a χ^2 worse than the nominal one by more than 25 on the same data, is not a plausible alternative and is excluded: this removes the single exponential ($\Delta\chi^2 = 134$, Sec. IV E). The largest remaining $|\Delta|$ of each group is taken as its uncertainty, since the variants of a group are

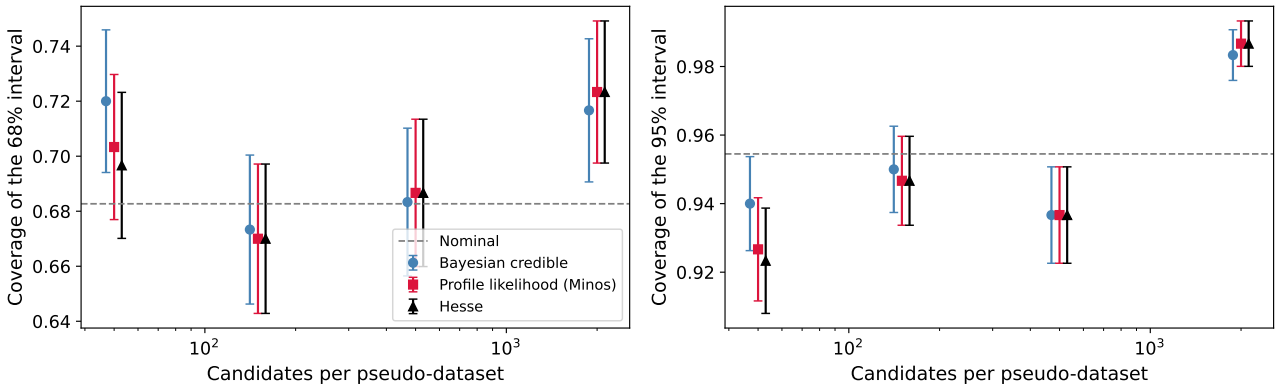


FIG. X. Coverage of the 68% (left) and 95% (right) intervals on A vs the number of candidates, for the Bayesian credible interval, the profile-likelihood (Minos) interval and the Hesse interval of the same likelihood; 300 pseudo-datasets per point, binomial uncertainties. The dashed lines mark the nominal coverage (1σ, 2σ; the Bayesian 95% interval is the 2.5–97.5% one).

alternative descriptions of the same choice, and the four groups are added in quadrature.

The results are in Table II, and all the Bayesian fits, with their intervals and goodness of fit, in Table V of the Appendix. The two methods give the same shifts to within ∼0.0005 and the same group uncertainties, with total systematic uncertainties of 0.0070 (Bayesian) and 0.0072 (frequentist). The final results are

$$\text{Bayesian:}$$
$$A_{\text{raw}} = -0.0443 \pm 0.0085\,(\text{stat}) \pm 0.0070\,(\text{syst}),$$
$$\text{frequentist:}$$
$$A_{\text{raw}} = -0.0441 \pm 0.0085\,(\text{stat}) \pm 0.0072\,(\text{syst}).$$

The preselection dominates, through the single variant `ProbK` > 0.6 ($+0.0058$, $1.2\,\sigma_B$). To tell a dependence on the PID requirement from a fluctuation of that single value, the frequentist fit was repeated in steps of 0.05 of each cut, from `ProbK` > 0.30 to > 0.85 and from `ProbPi` < 0.15 to < 0.70. On the tight side of the kaon requirement the shifts scatter around zero without a trend ($+0.0009$, $+0.0058$, -0.0017, -0.0112, -0.0054, $+0.0010$, $+0.0022$ for `ProbK` > 0.55–0.85, all within $1.3\,\sigma_B$), and the pion requirement gives a flat scan within $1.3\,\sigma_B$: the `ProbK` > 0.6 value is a statistical fluctuation, and the preselection uncertainty is a conservative estimate. Loosening the kaon requirement, instead, moves A_{raw} monotonically (-0.0019, -0.0032, -0.0067, -0.0111 for `ProbK` > 0.45–0.30, the last at $2.5\,\sigma_B$), as expected if misidentified $B \to K\pi\pi$ and $B \to \pi KK$ decays, with asymmetries of their own and not included in the mass model, enter the sample; this supports the nominal requirement. The combination rule was fixed before this scan and is not changed by it. The background group is the second contribution and is driven by the assumptions on the charge dependence of the background, the window the third.

Each method computes σ_B from its own uncertainties (posterior standard deviation, Hesse error). Since σ_B is the difference of two nearly equal uncertainties, it is sensitive to small differences between them, including the Monte Carlo noise of the posterior standard deviation (1–3%), and a variant close to the threshold can be classified differently by the two methods: `ProbPi` < 0.4 counts for the Bayesian fit ($|\Delta| = 0.00223$, $\sigma_B = 0.00220$) and not for the frequentist one (0.00204 vs 0.00229). This does not affect the result, since the uncertainty of each group is set by its largest shift, the same in both methods.

VI. MACHINE LEARNING MODULE

As a complementary, data-driven approach to the signal/background separation of the mass fits (Secs. III–IV), supervised classifiers were trained on weak labels derived from the mass spectrum, and used to study whether an ML-assisted event selection changes A_{raw} or the mass spectrum, and whether an adaptive partition of the Dalitz plane resolves local structure.

A. Weak Labels and Feature Set

No candidate carries a true signal/background label, so the classifiers are trained with weak labels, following the classification-without-labels (CWoLa) approach [16]: a classifier is trained to separate two *mixed* samples with different signal fractions, here a signal-enriched window around the B peak and the mass sidebands. If, within each class, the feature distributions are the same in the two samples, the optimal classifier between the two mixtures is a monotonic function of the signal-to-background likelihood ratio $p_s(x)/p_b(x)$, which by the Neyman-Pearson lemma is the most powerful test statistic for signal against background: the classifier trained on the mixtures is also the optimal signal/background classifier [16]. Two conditions follow. The background in the sidebands must be representative of the background

Group	Variant	n	Δ Bayes	σ_B Bayes	Δ freq.	σ_B freq.
Window	$[5050, 5600]$	21164	-0.0003	0.0009	-0.0004	0.0005
	$[5150, 5600]$	18566	-0.0001	0.0007	-0.0005	0.0021
	$[5100, 5450]$	19004	$+0.0002$	0.0009	-0.0001	0.0003
	$[5100, 5700]$	20191	$\mathbf{-0.0014}$	0.0010	$\mathbf{-0.0014}$	0.0001
Background	single exponential (rejected)	19731	-0.0037	–	-0.0039	–
	shape shared between charges	19731	$\mathbf{-0.0037}$	–	$\mathbf{-0.0040}$	–
	symmetric, $A_b = 0$	19731	-0.0028	–	-0.0031	–
Signal	Crystal Ball	19731	$\mathbf{-0.0002}$	–	$\mathbf{-0.0006}$	–
	μ, σ per charge	19731	-0.0000	–	-0.0001	–
Preselection	`ProbK` > 0.4	26553	-0.0032	0.0038	-0.0032	0.0037
	`ProbK` > 0.6	13981	$\mathbf{+0.0058}$	0.0048	$\mathbf{+0.0058}$	0.0050
	`ProbPi` < 0.4	17984	$+0.0022$	0.0022	$+0.0020$	0.0023
	`ProbPi` < 0.6	20953	-0.0004	0.0020	-0.0006	0.0017
Group uncertainties: window / background / signal / preselection				0.0014/0.0037/0.0002/0.0058		0.0014/0.0040/0.0006/0.0058
Total (quadrature)				0.0070		0.0072

TABLE II. Systematic study common to the Bayesian and frequentist fits: shift of A_{raw} for each variant and, for the variants that change the sample, the expected statistical spread σ_B of the shift. The largest contributing shift of each group is in bold. The single exponential is rejected by the data and excluded. For the window $[5150, 5600]$ the frequentist uncertainty is scaled from the number of events, since the Hesse matrix of that fit is not accurate.

under the peak; and the features must not be correlated with the B mass. The second condition cannot hold exactly: the reconstructed mass is not an input feature, but the nine momentum components of the three kaons determine it, so a flexible model can partly learn the mass, separate the two mass regions rather than signal from background, and *sculpt* the mass distribution of the candidates that pass a cut on its score. Section VI F tests this directly.

The labels use the nominal Bayesian fit (Sec. IV C, $\mu = 5284.39$, $\sigma = 18.00$ MeV). Candidates with $|m - \mu| < 2\sigma$, $m \in [5248.4, 5320.4]$ MeV, are labeled signal-like (15,130); candidates in $[5150, \mu - 4\sigma] \cup [\mu + 4\sigma, 5600] = [5150, 5212.4] \cup [5356.4, 5600]$ MeV are labeled background-like (1940); the 2–4σ gap is left unlabeled. The region 5100–5150 MeV, inside the fit window, is excluded from the background-like class to satisfy the first condition: there the background rises towards partially reconstructed B decays (Sec. II B), whose kinematics differ from those of the combinatorial background under the peak. The classes are imbalanced ($1940/15{,}130 = 0.128$). Every model is trained with the same balanced class weights, $w_c = N/(2N_c)$, i.e. 4.40 for background-like and 0.56 for signal-like candidates (through `class_weight` for the scikit-learn models and the network, and through sample weights for gradient boosting), so that the two classes contribute equally to the loss.

The feature set consists of the 6 per-track kinematic/PID variables (`PX`, `PY`, `PZ`, `ProbK`, `ProbPi`, `IPChi2`) for each of the 3 final-state tracks (18 features), plus two B-candidate variables not used in the preselection: `B_FlightDistance` and `B_VertexChi2`, standard discriminants against combinatorial background (a genuine B flies further before decaying and has a better-reconstructed vertex than a random track combination). Both separate the weak-label classes (Kolmogorov-Smirnov $p < 10^{-60}$) and are not redundant with the per-track features (Spearman $|\rho| \leq 0.53$ with each of the 18), giving 20 features. A principal-component analysis of the standardized features needs 15 of 20 components to reach 90% of the variance, so no dimensionality reduction is applied. The labeled sample was split 60/20/20 into training, validation and test sets (10,242/3414/3414 candidates, 388 background-like in the test set), stratified on the label, with the feature scaler fit on the training set only.

B. Model Comparison

Six classifiers were trained: logistic regression, without and with an $L1$ penalty (linear baselines), a decision tree (depth 4), a Random Forest [17] (300 trees of depth 8), gradient boosting [18] (200 stages of depth-3 trees, learning rate 0.05), and a feed-forward neural network (FFNN: two hidden layers of 32 and 16 units with ReLU activation, dropout 0.2, binary cross-entropy, the Adam optimizer [19] with learning rate 10^{-3}, early stopping on the validation AUC). The models are compared through the area under the receiver operating characteristic (ROC) curve (AUC), which equals the probability that a randomly chosen signal-like candidate receives a higher score than a randomly chosen background-like one (0.5 for a random classifier, 1 for a perfect one). Table III gives the 5-fold cross-validated AUC on the training set and the AUC on the untouched test set, with its bootstrap uncertainty. The reference model for the rest of the section was chosen on the cross-validated AUC, so that the test set is not used for any choice: gradient boosting (0.866), essentially tied with the Random Forest (0.863).

The three non-linear ensembles and the FFNN improve the AUC by $\sim$0.09 over the linear models and by $\sim$0.04

Model	CV AUC	Test AUC	Brier
Logistic Regression	0.769 ± 0.016	0.771 ± 0.013	0.197
Logistic Regression (L1)	–	0.771 ± 0.013	0.198
Decision Tree	0.816 ± 0.011	0.824 ± 0.013	0.133
Random Forest	0.863 ± 0.017	0.860 ± 0.012	0.122
Gradient Boosting	$\mathbf{0.866 \pm 0.016}$	0.864 ± 0.012	0.116
FFNN	–	0.857 ± 0.012	0.134

TABLE III. Cross-validated AUC (mean and standard deviation over the folds), test-set AUC (with bootstrap uncertainty) and test-set Brier score of the raw scores. Gradient boosting, best in cross-validation, is the reference model. The FFNN is validated on the validation set instead of by cross-validation.

over the single tree (Fig. XI), while their mutual differences (≤ 0.007) are smaller than the test-set uncertainty (0.012). The FFNN training is reproducible (fixed seeds, deterministic operations); its dependence on the random initialization, measured by retraining it with ten seeds, is 0.854 ± 0.005 in test AUC.

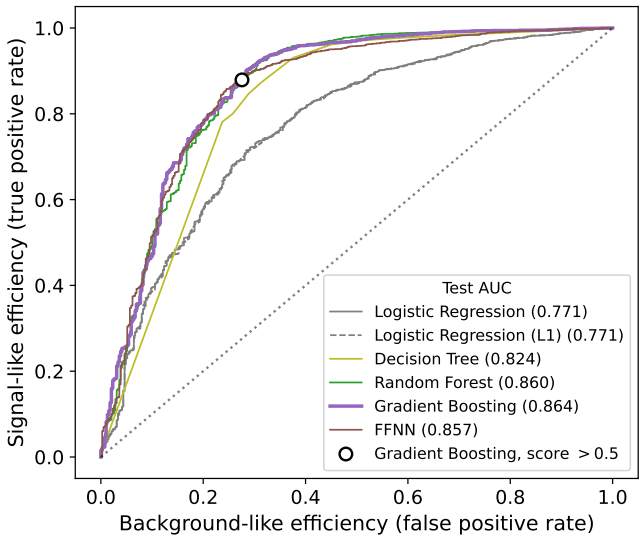


FIG. XI. ROC curves on the test set, with the test AUC of each model in the legend (Table III); the reference model is drawn thicker, and the two logistic regressions overlap. The circle marks the cut on the reference score used in Sec. VI F.

The two measures of the relevance of a feature available here – the Random Forest importance (the decrease in Gini impurity $G = 1 - \sum_c p_c^2$, p_c the class fractions in a node, produced by the splits on that feature) and the magnitude of the $L1$-regularized (Lasso [20]) logistic-regression coefficient of the standardized feature – agree only partly (Fig. XII; Spearman correlation of the two rankings 0.42, $p = 0.06$). The impact-parameter variables `IPChi2` dominate the Random Forest, while the Lasso coefficients are largest for `B_FlightDistance`, `H3_IPChi2`, `H2_PZ` and `B_VertexChi2`. The difference is expected: the Gini importance reflects the use of a feature in non-linear splits and interactions, and is shared among correlated features, while a linear coefficient measures the effect of a feature with the others held fixed. Both measures rank the two B-candidate variables among the top five (Random Forest 4th and 5th, Lasso 1st and 4th): they summarize in one number the displaced-vertex topology that the per-track variables encode only jointly.

C. Regularization, Bias-Variance, and Calibration

A scan of the regularization strength of the logistic regression (30 log-spaced values of $C = 1/\alpha$, cross-validated) gives the same optimal AUC for the $L1$ and $L2$ penalties (0.7686, at $C_{L1} = 0.12$ and $C_{L2} = 9.2$), and the optimal $L1$ penalty sets no coefficient to zero: the linear model does not overfit, and its limitation is the linear decision boundary, not the variance.

The bias-variance trade-off was studied with gradient boosting, whose complexity grows with the number of boosting stages (Fig. XIII). The test AUC rises up to $\sim$150 stages and then stays at 0.865, while the training AUC keeps rising: the train-test gap grows from 0.018 at 20 stages to 0.084 at 300. The additional stages fit fluctuations of the training sample without improving the ranking of new candidates – overfitting, which the default 200 stages of Table III do not reach in the test AUC. The accuracy at the 0.5 threshold stays below that of always predicting the majority class (0.886): with balanced class weights the classifier is trained to minimize the weighted error, in which a misclassified background-like candidate counts eight times more than a signal-like one, and not the unweighted error rate.

The AUC depends only on the ranking of the scores; whether a score can be read as a probability is measured by the Brier score [21], $\mathrm{BS} = \frac{1}{N}\sum_i (p_i - y_i)^2$, the mean squared difference between predicted probability p_i and label $y_i \in \{0, 1\}$ (lower is better). Class weights change the probabilities a model learns: a classifier trained with weights w_s, w_b on the two classes converges to

$$\tilde{p} = \frac{w_s p}{w_s p + w_b (1 - p)}$$

instead of the true p, i.e. it describes a world in which the classes are re-balanced. With $w_b > w_s$ the raw scores therefore *underestimate* the probability of the signal-like class, which is what the reliability diagram of the Random Forest shows (Fig. XIV): at a raw score of 0.25 the observed signal-like fraction is already 0.58. An isotonic recalibration [22], fitted on held-out folds of the unweighted training sample, follows the diagonal and halves the Brier score, from 0.122 to 0.062. Most of the gain undoes the re-weighting rather than correcting the model; the ranking, and hence the AUC and any selection based on a score threshold, is unchanged by a monotonic recalibration.

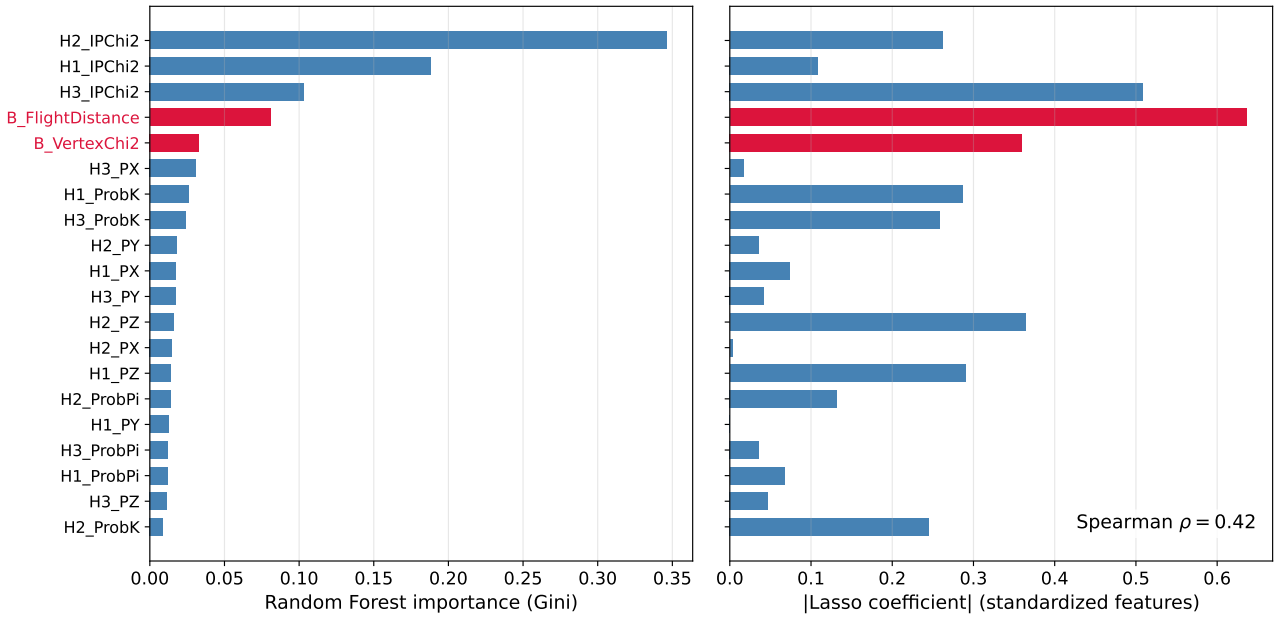


FIG. XII. Random Forest feature importance (Gini, left) and magnitude of the Lasso coefficient of the standardized feature (right) for the 20 features, both ordered by Random Forest importance; the two B-candidate variables are highlighted.

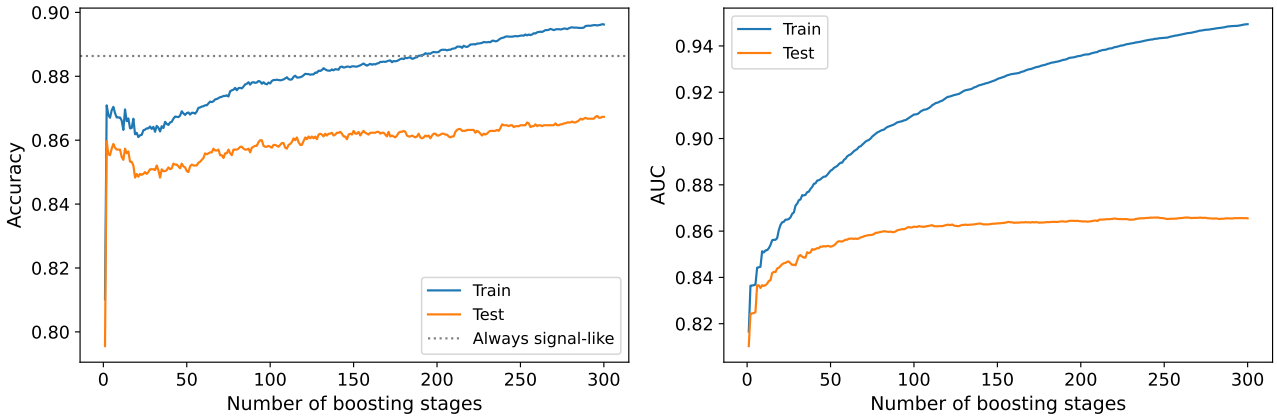


FIG. XIII. Gradient boosting: accuracy at the 0.5 threshold (left) and AUC (right) on the training and test sets as a function of the number of boosting stages. The dotted line is the accuracy of always predicting the signal-like class.

D. Uncertainty Quantification

The uncertainty of a prediction has two components: *aleatoric*, from the intrinsic overlap of the classes in feature space, which more training data cannot reduce, and *epistemic*, from the limited knowledge of the model parameters, which more training data reduces. Three estimators were compared, each giving an ensemble of M predictions $p_m(x)$ for each candidate and quoting their spread, $\sigma(x) = \left[\frac{1}{M}\sum_m (p_m(x) - \bar{p}(x))^2\right]^{1/2}$: the $M = 300$ trees of the Random Forest, each trained on a bootstrap sample; $M = 300$ forward passes of the FFNN with dropout kept active at inference (MC dropout), which approximates sampling from a Bayesian posterior over the network weights [23]; and a deep ensemble [24] of $M = 10$ FFNNs trained from different random initializations. The mean per-candidate uncertainty is 0.178 for the Random Forest, 0.095 for MC dropout (0.099 ± 0.006 over the ten initializations) and 0.050 for the deep ensemble (Fig. XV), with the same values within 10% for background-like and signal-like candidates.

The three estimators answer different questions, which explains the factor ∼3.5 between them. The deep ensemble only varies the initialization, on the same training data; MC dropout also perturbs the network structure; the trees of the Random Forest differ in their training data (bootstrap) and in the features they use, and each tree, of limited depth, is individually a poor and discrete estimator of p. Their per-candidate values are cor-

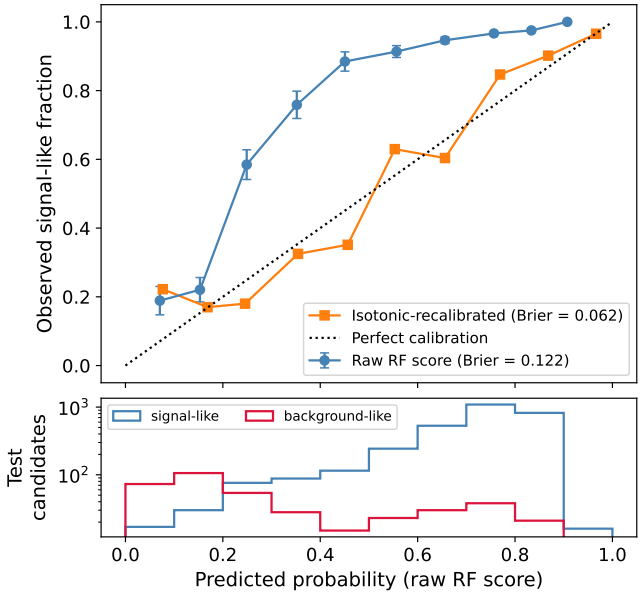


FIG. XIV. Reliability diagram of the Random Forest on the test set: observed fraction of signal-like candidates vs. predicted probability, for the raw score (with binomial uncertainties) and for the isotonic-recalibrated one. Bottom: distribution of the raw score for the two weak-label classes; the low-probability bins contain few candidates.

related but not interchangeable (Spearman correlations 0.41–0.57 between pairs). The Random Forest spread contains both components. It is largest for scores around 0.5 (Fig. XV, left), where the classes overlap and the trees disagree most – an aleatoric contribution; and it decreases with the size of the training set, from 0.220 with a quarter of it to 0.204 with half and 0.179 with the full set, while the test AUC stays at 0.860–0.862 – the epistemic contribution, which more data reduces.

E. K-means on the Dalitz Plot

K-means is the unsupervised part of the module: it uses no labels, and the partition of the Dalitz plane is learned from the density of the candidates rather than fixed by hand. As an adaptive alternative to the fixed Dalitz-plot binning of Sec. IV H, the candidates of the fit window were partitioned with K-means [25] on the two standardized Dalitz variables $(m^2_{\rm lo}, m^2_{\rm hi})$. K-means partitions the points x_i into K clusters C_k minimizing the within-cluster sum of squares (inertia) $W = \sum_k \sum_{i\in C_k} \|x_i - \bar{x}_k\|^2$, with $\bar{x}_k$ the cluster centroids; K has to be fixed in advance. The quality of a partition is measured by the silhouette [26]: for each point, $s_i = (b_i - a_i)/\max(a_i, b_i)$ compares its mean distance a_i to the other points of its cluster with the mean distance b_i to the points of the nearest other cluster, and the average over all points ($-1 \le s \le 1$, higher is better) measures how well separated the clusters are. $K = 8$ was fixed in advance, as a compromise between resolution and the number of candidates per cluster (at least 1233), and $K = 3$, the silhouette optimum over $K = 2$–12 (0.46, against 0.41 for $K = 8$, a local maximum), was added. Both partitions are stable: repeating the clustering with a different random initialization gives an adjusted Rand index [27] above 0.998.

The asymmetry of each cluster was compared with the global one as in Sec. IV H, $S_k = (A_k - A_{\rm glob})/\sigma(A_k)$. For $K = 8$ (Fig. XVI), two clusters deviate by more than 3σ (probability 2×10^{-4} for two or more if the asymmetry were uniform), and the homogeneity test gives $\chi^2 = 46.4/7$ ($p = 7 \times 10^{-8}$). The most significant one lies at low $m^2_{\rm lo}$ (1–2.5 GeV^2) and intermediate $m^2_{\rm hi}$, with $A_k = -0.127 \pm 0.017$ ($n = 3601$, $S_k = -5.1$); two clusters at high $m^2_{\rm hi}$ have positive asymmetries, $A_k = +0.020 \pm 0.020$ ($S_k = +3.2$) and $+0.040 \pm 0.028$ ($+2.9$). In the signal window ($|m - \mu| < 2\sigma$) the same pattern is sharper – three clusters beyond 3σ, with $S_k = -5.6$, $+4.4$ and $+3.0$, and $\chi^2 = 62.3/7$ – so it is not produced by the background. The coarser $K = 3$ partition gives $\chi^2 = 15.2/2$ ($p = 5\times10^{-4}$). The adaptive partition therefore confirms the picture of the fixed binning, a negative local asymmetry at low $m^2_{\rm lo}$ and positive ones at high $m^2_{\rm hi}$, with the same caveats (no background subtraction, no correction for detection asymmetries).

F. Stability of $A_{\rm raw}$ under ML Selection

The reference (gradient-boosting) score was applied to all candidates of the fit window (on the test set, a cut at 0.5 keeps 88% of the signal-like and 28% of the background-like candidates, Fig. XI), not only to the labeled ones used in training, and the counting asymmetry was recomputed after cuts on the score from 0.1 to 0.9 (Fig. XVII). Since every selected sample is contained in the full one, the expected statistical spread of the difference is $\sqrt{\sigma^2_{\rm sub} - \sigma^2_{\rm full}}$, not the uncertainty of either estimate. Over all thresholds, which keep from 18,443 to 2621 candidates, the shift with respect to the full window ($A_{\rm count} = -0.0432$) stays within 1.4 times this spread: the selection does not introduce a detectable bias in the counting asymmetry. Split by magnet polarity at a cut of 0.5, the counting asymmetries are -0.060 ± 0.010 (down) and -0.024 ± 0.012 (up), 2.2σ apart; the frequentist fit of the same samples, which separates the background, gives -0.054 ± 0.011 and -0.032 ± 0.013, 1.3σ apart (against 0.7σ for the full sample, Sec. IV F), so most of the larger difference of the counting estimates comes from the background.

Counting cannot reveal whether the selection deforms the mass spectrum. To test this, the selected sample (score > 0.5, 15,709 candidates) was refitted with the nominal mass model by both inference methods of this work, the Bayesian fit of Sec. IV and the frequentist maximum-likelihood fit of Sec. III, and the three estimates were compared with those of the full sample (Ta-

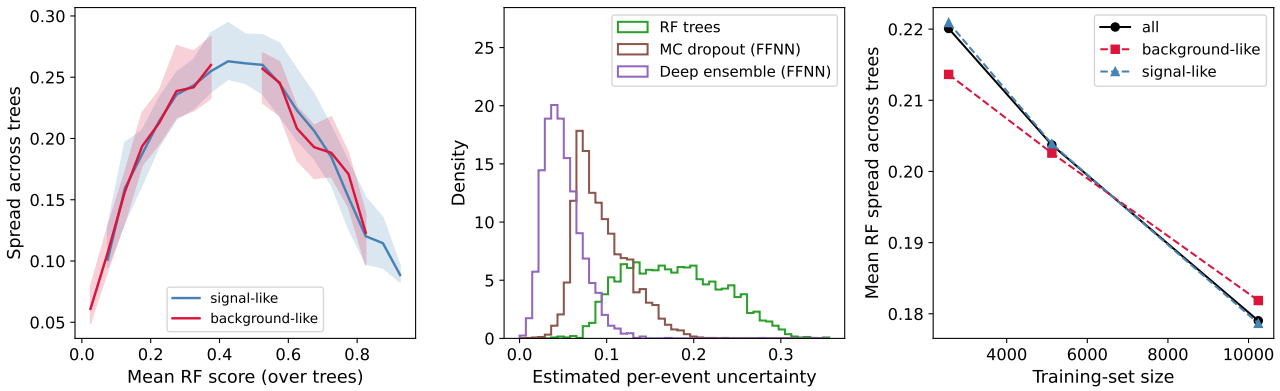


FIG. XV. Left: spread across the Random Forest trees vs. mean tree score, median and 16–84% band of the test candidates of each weak-label class. Center: distributions of the per-candidate uncertainty from the Random Forest trees, MC dropout and the deep ensemble. Right: mean Random Forest spread vs. training-set size, for all, background-like and signal-like test candidates.

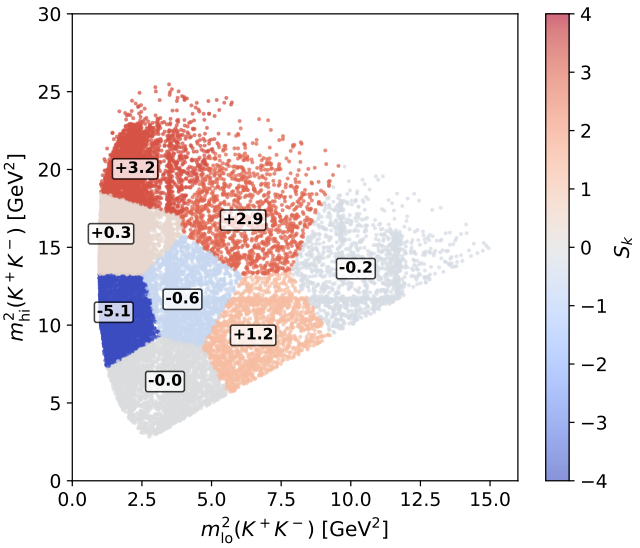


FIG. XVI. K-means partition ($K = 8$) of the Dalitz plane, all candidates of the fit window, same axes as Fig. VIII. Color and labels: local significance of each cluster with respect to the global asymmetry, $S_k = (A_k - A_{\rm glob})/\sigma(A_k)$, on the same scale as the right panel of Fig. VIII.

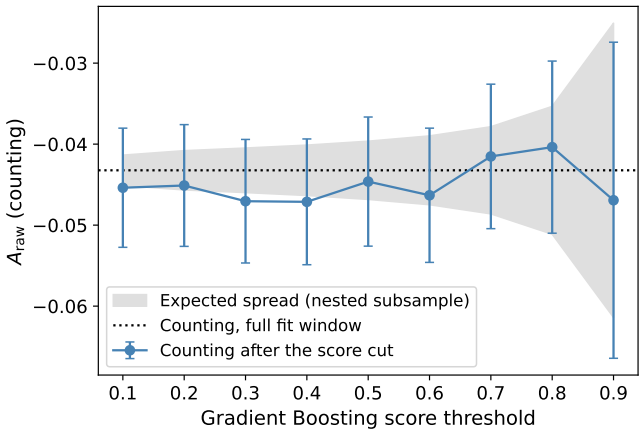


FIG. XVII. Counting asymmetry after a cut on the gradient-boosting score, vs. threshold, with its binomial uncertainty. Dotted line: counting asymmetry of the full fit window; band: expected statistical spread of a nested subsample around it, $\pm\sqrt{\sigma^2_{\rm sub} - \sigma^2_{\rm full}}$.

ble IV, Fig. XVIII). The asymmetry is stable: all three shifts are within their expected nested-sample spread, and the significance of $A \neq 0$ from the profile likelihood (Wilks' theorem [8]) is 5.2σ in both samples. The selection keeps 92% of the fitted signal and 36% of the background ($N_s = 14{,}121$, $N_b = 1590$), but this does not improve the precision on $A_{\rm raw}$ (0.0087 against 0.0085): the background under the peak is already small (≈4% in the signal window) and the fit separates it from the signal anyway. The mass spectrum, instead, changes. The goodness of fit of the nominal model degrades from $\chi^2 = 164/88$ to $253/88$ in the Bayesian fit and from $442/388$ to $557/388$ in the frequentist one (with a different binning), and the background shape changes: the slope of the slow exponential grows from 0.0028 to 0.0071 MeV^{-1} (B^+) and from 0.0016 to 0.0057 MeV^{-1} (B^-), and the fraction of the steep one from 0.11 to 0.28 and from 0.21 to 0.50. Two mechanisms can produce this: the classifier could reconstruct the mass from the kaon momenta and sculpt the background (Sec. VI A), or the selection could change the composition of the background. The first is excluded: the same classifier trained without the nine momentum components (test AUC 0.860) gives the same degradation ($\chi^2 = 533/388$). The efficiency of the cut as a function of the mass shows the second (Fig. XIX): it is 0.89–0.90 at the peak, ≈ 0.70 below 5150 MeV and 0.06–0.13 above 5375 MeV. The classifier does what it is trained for – it removes the combinatorial background and keeps genuine B decays, including the partially re-

constructed ones and the other non-combinatorial components that the weak labels cannot separate from the signal. In the selected sample these components weigh relatively more, and the nominal model, which has no dedicated component for them, describes it worse, mostly under the peak (the χ^2 of the $\pm 4\sigma$ region grows from 75 to 134 for 30 bins) and below it. A Crystal Ball signal gives the same degradation (407 → 522 for 386 degrees of freedom), so it is not due to the signal tails. A_{raw} is unaffected because it is determined by the peak and the background shape is free in the fit, but a measurement that relied on the background shape, or on a background model taken from the sidebands of the full sample, would be biased by the selection.

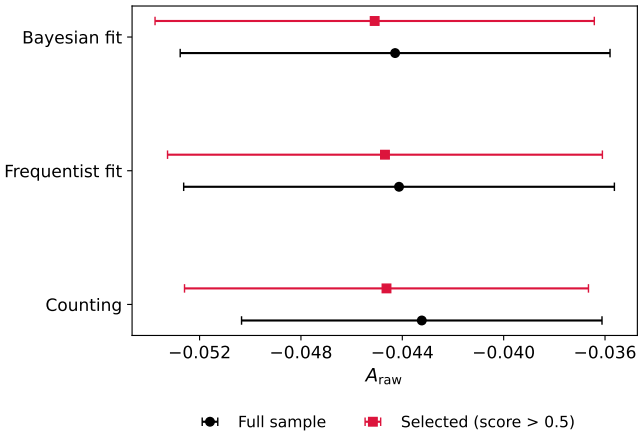


FIG. XVIII. A_{raw} from the Bayesian fit, the frequentist fit and counting, for the full sample and the ML-selected one (Table IV).

G. Hyperparameter Optimization

The hyperparameters of the Random Forest (120 configurations), gradient boosting (100) and the FFNN (168: seven architectures, four dropout rates, six learning rates) were optimized by repeated cross-validation on the training set (Tables VI–VII). The best configurations improve the test AUC over the defaults of Table III by 0.005 (Random Forest), 0.001 (gradient boosting) and 0.004 (FFNN), all well below the test-set uncertainty (0.012) and, for the FFNN, comparable with its spread over initializations (0.005). For the two tree ensembles the best cross-validated AUC is higher than the test AUC of the same configuration (0.871 vs 0.866 and 0.868 vs 0.865), as expected when the maximum is taken over many noisy cross-validation estimates (the "winner's curse"): the selection favors configurations with a favorable fluctuation, and the cross-validated score of the selected configuration is an optimistic estimate of its performance. The searches show that the defaults are already close to optimal for this problem, so the defaults are used throughout.

VII. COMPARISON AND DISCUSSION

[TODO: Shared, written last once §III exists – the one section that genuinely can't be finished without Marco. Headline comparison table: A_{raw} (counting, frequentist, Bayesian, ML-assisted) with uncertainties, side by side. Comparison with the published A_{CP}(KKK) value from Ref. [1] and discussion of why $A_{\text{raw}} \neq A_{CP}$. Local Dalitz-plot asymmetry: fixed binning vs K-means, and whether the frequentist local significance study (if Marco does one) agrees.]

VIII. LIMITATIONS AND FUTURE WORK

Several modeling choices and known weaknesses limit how far the results of Secs. IV–VI can be pushed, and are collected here rather than scattered through the results sections.

On the mass fits (Secs. IV–V): no tested signal model reaches $\chi^2/\text{ndof} \approx 1$ (the best, a Crystal Ball, gives 131.0/86, Sec. IV E), although A_{raw} does not depend on the signal shape; the systematic uncertainty, 0.007, is almost as large as the statistical one; it is dominated by a PID variant that a finer scan identifies as a statistical fluctuation (Sec. V), so it is conservative, but the combination rule cannot separate the two; the Savage-Dickey Bayes factor of Sec. IV D relies on a Gaussian approximation of the posterior far in its tail and depends on the prior width, though the conclusion (decisive evidence against $A = 0$) is robust; the toy-MC coverage check of Sec. IV G is a closure test and cannot detect model misspecification; and the local Dalitz asymmetries (Secs. IV H and VI E) use a coarse partition, a mass-window cut rather than a proper background subtraction, and no correction for momentum-dependent detection asymmetries.

On the machine-learning side (Sec. VI): the weak labels are imbalanced (1940 background-like vs. 15,130 signal-like candidates, Sec. VI A), and the test set contains only 388 background-like candidates, so differences in test AUC below ∼0.01 cannot be resolved; the three uncertainty estimators differ by a factor ∼3.5 in scale and measure different sources of variability (Sec. VI D); and the score-based selection changes the composition of the background, keeping the partially reconstructed and other non-combinatorial B decays that the nominal model does not describe (Sec. VI F); removing the momentum features does not help, since the weak labels cannot separate these decays from the signal.

More generally, no detector-level systematic correction (selection efficiency, production asymmetry) is applied anywhere in this work, so A_{raw} as reported throughout is explicitly *not* a measurement of A_{CP} (Sec. VII). The published analysis corrects for $A_P + A_D$ with $B^\pm \to J/\psi K^\pm$, $J/\psi \to \mu^+\mu^-$ [1], which is not part of this record; the $J/\psi \to K^+K^-$ decays inside our sample cannot replace

Method	Full sample	Score > 0.5	Shift (expected)
Bayesian fit	-0.0443 ± 0.0085	-0.0451 ± 0.0087	$-0.0008\,(0.0018)$
Frequentist fit	-0.0441 ± 0.0085	-0.0447 ± 0.0086	$-0.0006\,(0.0012)$
Counting	-0.0432 ± 0.0071	-0.0446 ± 0.0080	$-0.0014\,(0.0036)$

TABLE IV. A_{raw} on the full fit window (19,731 candidates) and after the cut on the gradient-boosting score (15,709), by the three methods. The expected shift is the statistical spread of a nested subsample, $\sqrt{\sigma^2_{\text{sub}} - \sigma^2_{\text{full}}}$. Counting includes the background.

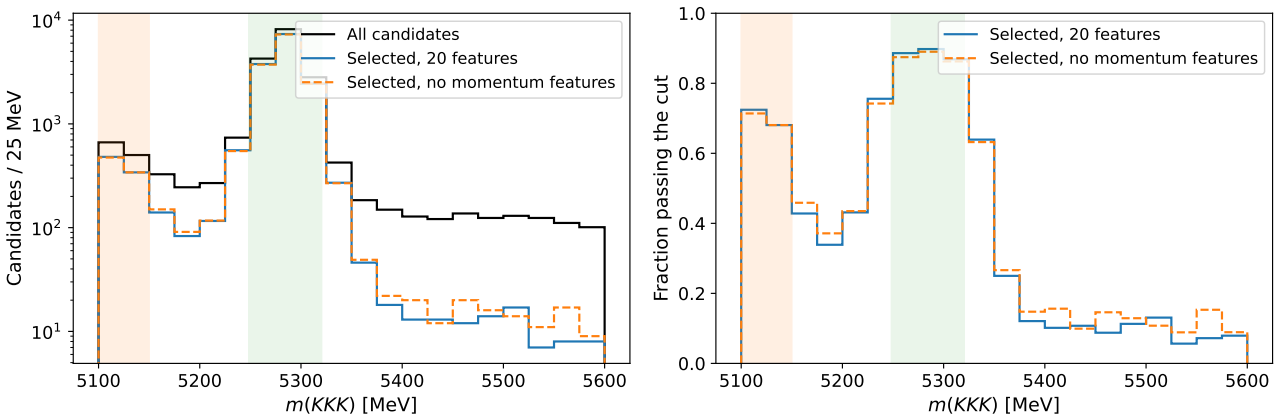


FIG. XIX. Left: mass spectrum of all candidates and of those with gradient-boosting score > 0.5, for the 20-feature classifier and for one trained without the kaon momentum components (25 MeV bins). Right: fraction of candidates passing the cut. Shaded: the region below 5150 MeV excluded from the background-like labels, and the signal-like window.

it, since this mode is rare and the 568 candidates under the B peak in the J/ψ band are dominated by charmless decays ($A_{\text{count}} = -0.165 \pm 0.041$, a local Dalitz-plot asymmetry rather than a control measurement).

The most natural next steps, in rough priority order, are: correcting A_{raw} for the production and detection asymmetries with a control channel such as $B^{\pm} \to J/\psi K^{\pm}$, $J/\psi \to \mu^+\mu^-$; replacing the fixed-cut local Dalitz asymmetries (both binned, Sec. IV H, and K-means, Sec. VI E) with a proper sWeight-based background subtraction; and an explicit model of the partially reconstructed and other non-combinatorial B backgrounds, which would also make an ML-selected sample usable for a fit.

IX. CONCLUSIONS

[TODO: Shared. Short – synthesize ONE key message per method (does not need to restate every number from the Results sections), state what was learned methodologically (e.g. "a minimal Gaussian+exponential model is insufficient, a Crystal Ball with floated shape parameters improves the fit without moving A_{raw}"), not just what was measured.]

X. AUTHOR CONTRIBUTIONS

Marco Belotti led the frequentist analysis (Sec. III): mass model, likelihood fits, and profile-likelihood confidence interval. Giacomo Moretti led the Bayesian analysis (Sec. IV): model definition, MCMC sampling, convergence diagnostics, Bayes factor, and the associated systematic/goodness-of-fit studies. The common nominal model, the systematic study (Sec. V), the machine learning analysis and the comparison of the three methods on the ML-selected sample (Sec. VI), the toy-MC validation, the Dalitz-plot analyses, and the final cross-method comparison (Sec. VII) were carried out jointly. Both authors contributed to the manuscript.

XI. CODE AND DATA AVAILABILITY

All code is written in Python $\geq$3.12, with `scikit-learn` [28] for the classical machine-learning models and `TensorFlow/Keras` [29] for the neural networks. Key package versions: `uproot 5.7`, `iminuit 2.32.0`, `emcee 3.1.6`, `corner 2.3.0`, `scikit-learn 1.9`, `tensorflow 2.21.0`, `keras 3.15.1`, `scipy 1.18`, `numpy 2.4`, `pandas 3.0`.

The dataset is the CERN Open Data record 4900 ("Matter Antimatter Differences – B meson decays to three hadrons"). The analysis code is available in the accompanying repository, in

`Code_general/Frequentist/` (frequentist fits and the common systematic study), `Code_general/Bayesian/` (nominal model, its checks, and the minimal model), and `Code_general/Machine_Learning/`; the figures of this paper are produced from the saved results of those analyses by the scripts in `Report/figures/` (`regen_mass_spectrum.py`, `regen_figures_nominal.py`, `regen_figures_ml_nominal.py`).

Appendix A: Full results tables

[TODO: Shared. Still needed: frequentist fit results alongside the Bayesian rows below, for a direct side-by-side comparison.]

[1] R. Aaij *et al.* (LHCb), Direct CP violation in charmless three-body decays of $B^{\pm}$ mesons, Phys. Rev. D **108**, 012008 (2023), arXiv:2206.07622 [hep-ex].

[2] L. Falcão, CP violation measurements in three-body charmless B decays, on behalf of the LHCb Collaboration, 12th Workshop on the CKM Unitarity Triangle (2024), arXiv:2403.19921 [hep-ex].

[3] LHCb Collaboration, R. Aaij, *et al.*, Measurement of CP violation in the phase space of $B^{\pm} \to K^{\pm}\pi^{+}\pi^{-}$ and $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decays, Physical Review Letters **111**, 101801 (2013), arXiv:1306.1246.

[4] LHCb Collaboration, Matter Antimatter Differences – B meson decays to three hadrons (2017), record 4900.

[5] LHCb Collaboration, Matter Antimatter Differences – Project Notebook (2017), record 4902.

[6] H. Dembinski, P. Ongmongkolkul, *et al.*, scikit-hep/iminuit, Zenodo (2020).

[7] F. James and M. Roos, Minuit – a system for function minimization and analysis of the parameter errors and correlations, Computer Physics Communications **10**, 343 (1975).

[8] S. S. Wilks, The large-sample distribution of the likelihood ratio for testing composite hypotheses, Annals of Mathematical Statistics **9**, 60 (1938).

[9] J. Goodman and J. Weare, Ensemble samplers with affine invariance, Commun. Appl. Math. Comput. Sci. **5**, 65 (2010).

[10] D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman, emcee: The MCMC Hammer, Publ. Astron. Soc. Pac. **125**, 306 (2013), arXiv:1202.3665.

[11] J. M. Dickey, The weighted likelihood ratio, linear hypotheses on normal location parameters, Ann. Math. Statist. **42**, 204 (1971).

[12] H. Jeffreys, *Theory of Probability*, 3rd ed. (Oxford University Press, Oxford, 1961).

[13] T. Skwarnicki, *A study of the radiative cascade transitions between the Υ' and Υ resonances*, Ph.D. thesis, Institute of Nuclear Physics, Krakow (1986), dESY-F31-86-02.

[14] A. Gelman, X.-L. Meng, and H. Stern, Posterior predictive assessment of model fitness via realized discrepancies, Statistica Sinica **6**, 733 (1996).

[15] R. Barlow, Systematic errors: facts and fictions, in *Conference on Advanced Statistical Techniques in Particle Physics* (Durham, 2002) arXiv:hep-ex/0207026.

[16] E. M. Metodiev, B. Nachman, and J. Thaler, Classification without labels: Learning from mixed samples in high energy physics, JHEP **10**, 174, arXiv:1708.02949.

[17] L. Breiman, Random forests, Machine Learning **45**, 5 (2001).

[18] J. H. Friedman, Greedy function approximation: a gradient boosting machine, Annals of Statistics **29**, 1189 (2001).

[19] D. P. Kingma and J. Ba, Adam: A method for stochastic optimization, in *3rd International Conference on Learning Representations (ICLR)* (2015) arXiv:1412.6980.

[20] R. Tibshirani, Regression shrinkage and selection via the lasso, Journal of the Royal Statistical Society B **58**, 267 (1996).

[21] G. W. Brier, Verification of forecasts expressed in terms of probability, Mon. Weather Rev. **78**, 1 (1950).

[22] B. Zadrozny and C. Elkan, Transforming classifier scores into accurate multiclass probability estimates, in *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (2002) pp. 694–699.

[23] Y. Gal and Z. Ghahramani, Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning, in *Proceedings of the 33rd International Conference on Machine Learning (ICML)* (2016) arXiv:1506.02142.

[24] B. Lakshminarayanan, A. Pritzel, and C. Blundell, Simple and scalable predictive uncertainty estimation using deep ensembles, in *Advances in Neural Information Processing Systems 30* (2017) arXiv:1612.01474.

[25] S. P. Lloyd, Least squares quantization in PCM, IEEE Transactions on Information Theory **28**, 129 (1982).

[26] P. J. Rousseeuw, Silhouettes: A graphical aid to the interpretation and validation of cluster analysis, J. Comput. Appl. Math. **20**, 53 (1987).

[27] L. Hubert and P. Arabie, Comparing partitions, Journal of Classification **2**, 193 (1985).

[28] F. Pedregosa *et al.*, Scikit-learn: Machine learning in Python, Journal of Machine Learning Research **12**, 2825 (2011).

[29] M. Abadi *et al.*, TensorFlow: Large-scale machine learning on heterogeneous distributed systems (2016), arXiv:1603.04467.

Fit	n	A_{raw} (median)	68% CI	χ^2/ndof
Nominal	19731	-0.0443	$[-0.0528, -0.0358]$	164.1/88
Systematic variants (Sec. V)				
Window $[5050, 5600]$	21164	-0.0446	$[-0.0531, -0.0362]$	217.4/98
Window $[5150, 5600]$	18566	-0.0444	$[-0.0528, -0.0359]$	141.5/78
Window $[5100, 5450]$	19004	-0.0440	$[-0.0524, -0.0357]$	136.9/58
Window $[5100, 5700]$	20191	-0.0456	$[-0.0540, -0.0373]$	200.2/108
Single exponential	19731	-0.0480	$[-0.0564, -0.0395]$	297.7/92
Background shape shared	19731	-0.0480	$[-0.0564, -0.0397]$	174.1/91
Symmetric background	19731	-0.0471	$[-0.0552, -0.0388]$	169.8/89
Crystal Ball signal	19731	-0.0445	$[-0.0529, -0.0363]$	131.0/86
μ, σ per charge	19731	-0.0443	$[-0.0527, -0.0356]$	164.2/86
`ProbK` > 0.4	26553	-0.0474	$[-0.0550, -0.0399]$	175.6/88
`ProbK` > 0.6	13981	-0.0385	$[-0.0482, -0.0289]$	145.0/88
`ProbPi` < 0.4	17984	-0.0421	$[-0.0506, -0.0333]$	150.1/88
`ProbPi` < 0.6	20953	-0.0446	$[-0.0529, -0.0366]$	159.0/88
Cross-checks (Secs. IV B, IV F)				
Dispersed start, run 1	19731	-0.0446	$[-0.0529, -0.0360]$	164.1/88
Dispersed start, run 2	19731	-0.0442	$[-0.0528, -0.0356]$	164.1/88
Magnet down	11636	-0.0497	$[-0.0607, -0.0386]$	153.1/88
Magnet up	8095	-0.0378	$[-0.0507, -0.0251]$	150.3/88
Narrow prior, $A \in (-0.3, 0.3)$	19731	-0.0443	$[-0.0528, -0.0358]$	164.1/88
D^0 veto	18744	-0.0496	$[-0.0584, -0.0409]$	159.1/88
D^0, J/ψ, χ_{c0} vetoes	16367	-0.0423	$[-0.0516, -0.0329]$	140.1/88

TABLE V. All Bayesian fits of the nominal model and of its variants: number of candidates, posterior median and 68% credible interval of A_{raw}, and binned goodness of fit (10 MeV bins). Every fit uses 32 walkers and 25,000 steps.

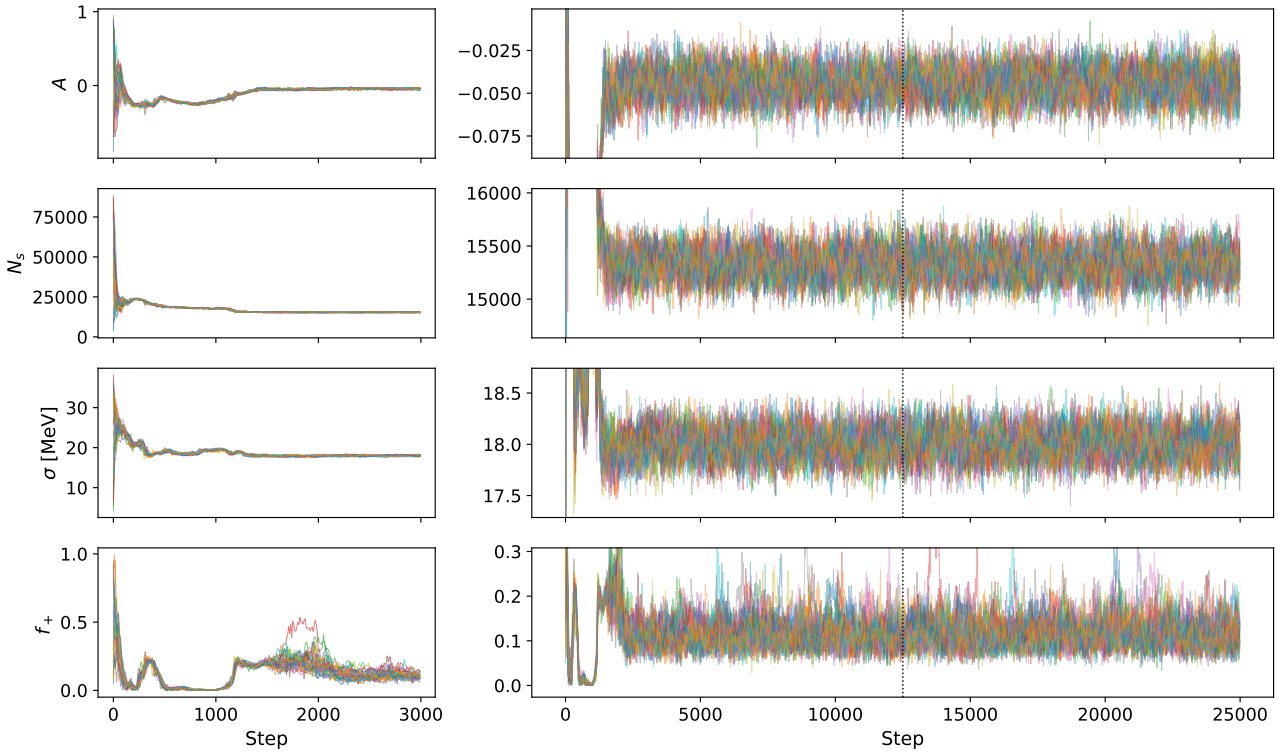


FIG. XX. Trace plot of the nominal fit with the 32 walkers initialized uniformly over the prior box (one line per walker, every 10th step), for A, N_s, σ and the steep-background fraction f_+. Left: first 3000 steps, full range; right: whole chain, zoomed on the posterior. The first half of the chain (dotted line) is discarded as burn-in.

Model	Grid	Best configuration	CV AUC / Test AUC (default)
Random Forest	`n_estimators` 200–1200, `max_depth` 6–12 or none, `min_samples_leaf` 5–40	400, 12, 5	0.8706 / 0.8656 (0.8604)
Gradient Boosting	`n_estimators` 100–500, `max_depth` 1–5, `learning_rate` 0.01–0.1	300, 2, 0.1	0.8684 / 0.8650 (0.8643)

TABLE VI. Hyperparameter search for the tree ensembles (120 and 100 configurations, 5-fold cross-validation repeated 10 times, balanced class weights). The test AUC of the best configuration is compared with that of the default of Table III (in parentheses); the bootstrap uncertainty of each test AUC is 0.012.

Architecture	Layers	Dropout	Learning rate	CV AUC
`deep_3layer` (best)	$[64, 32, 16]$	0.2	0.003	0.8585 ± 0.0174
`wide_2layer`	$[64, 32]$	0.3	0.003	0.8582 ± 0.0175
`deep_3layer`	$[64, 32, 16]$	0.3	0.003	0.8580 ± 0.0187
`wide_2layer`	$[64, 32]$	0.3	0.005	0.8579 ± 0.0184
`wide_2layer`	$[64, 32]$	0.4	0.005	0.8579 ± 0.0187

TABLE VII. Top five of the 168 FFNN configurations (7 architectures from one hidden layer of 8 units to five layers $[128, 64, 32, 16, 8]$, dropout 0.1–0.4, learning rates 0.0002–0.005; 5-fold cross-validation repeated twice). The five are within 0.0007 of each other, far below the fold-to-fold spread: the optimum is a plateau. The best configuration has test AUC 0.8606, against 0.8568 for the default architecture (± 0.005 over initializations).